Introduction

- If this stress and the restrained strain are allowed to develop to such an extent that they exceed the strength or strain capacity of concrete, then cracks will take place.

- Generally, 1/16 to 1/4-inch-wide cracks is acceptable limits (if its not leaking water).

- A 100-foot-long regular-weight concrete slab normally would shrink by about 3/4 inch. In other words, you should expect cracks totaling in widths up to 3/4 inch in every 100 feet of concrete.

About CRACKS

Cracks in buildings are common occurrence. It develops when the stress in a component exceeds the strength.

Classification of Cracks

Structural crack	Non Structural crack
○ Incorrect design ○ Faulty construction ○ Overloading	○ Internal induced stress in building material

Classification of Cracks

Based on width

Type	Width
Thin Medium Wide	< 1 mm 1-2 mm > 2 mm

Based on sight of crack

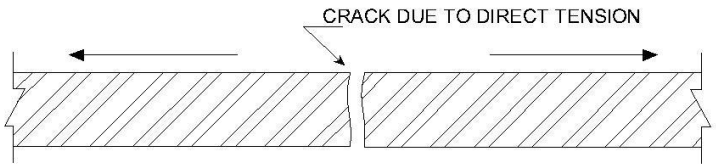
Vertical Horizontal Diagonal	Straight Toothed Stepped Map pattern Random	Uniform throughout Narrow at one end and gradually widening at the other

Common sight of crack

Vertical Horizontal Diagonal	Straight Toothed Stepped Map pattern Random	Uniform throughout Narrow at one end and gradually widening at the other

Internal stress in Building component

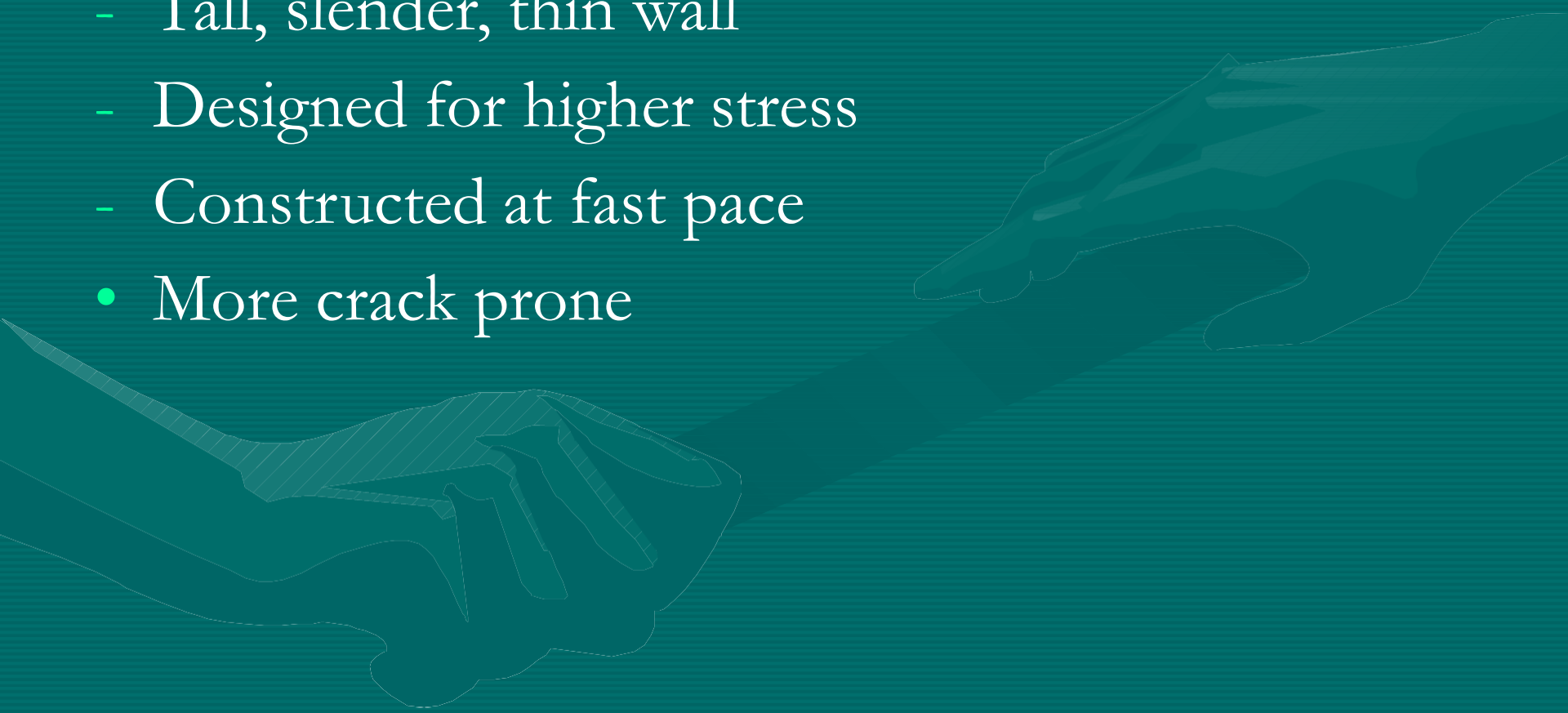
- Compressive
- Tensile
- Shear
- Building material
- 1. Masonry, Concrete, Mortar
  2. Weak in tension/shear
  3. Causing tension/shear crack

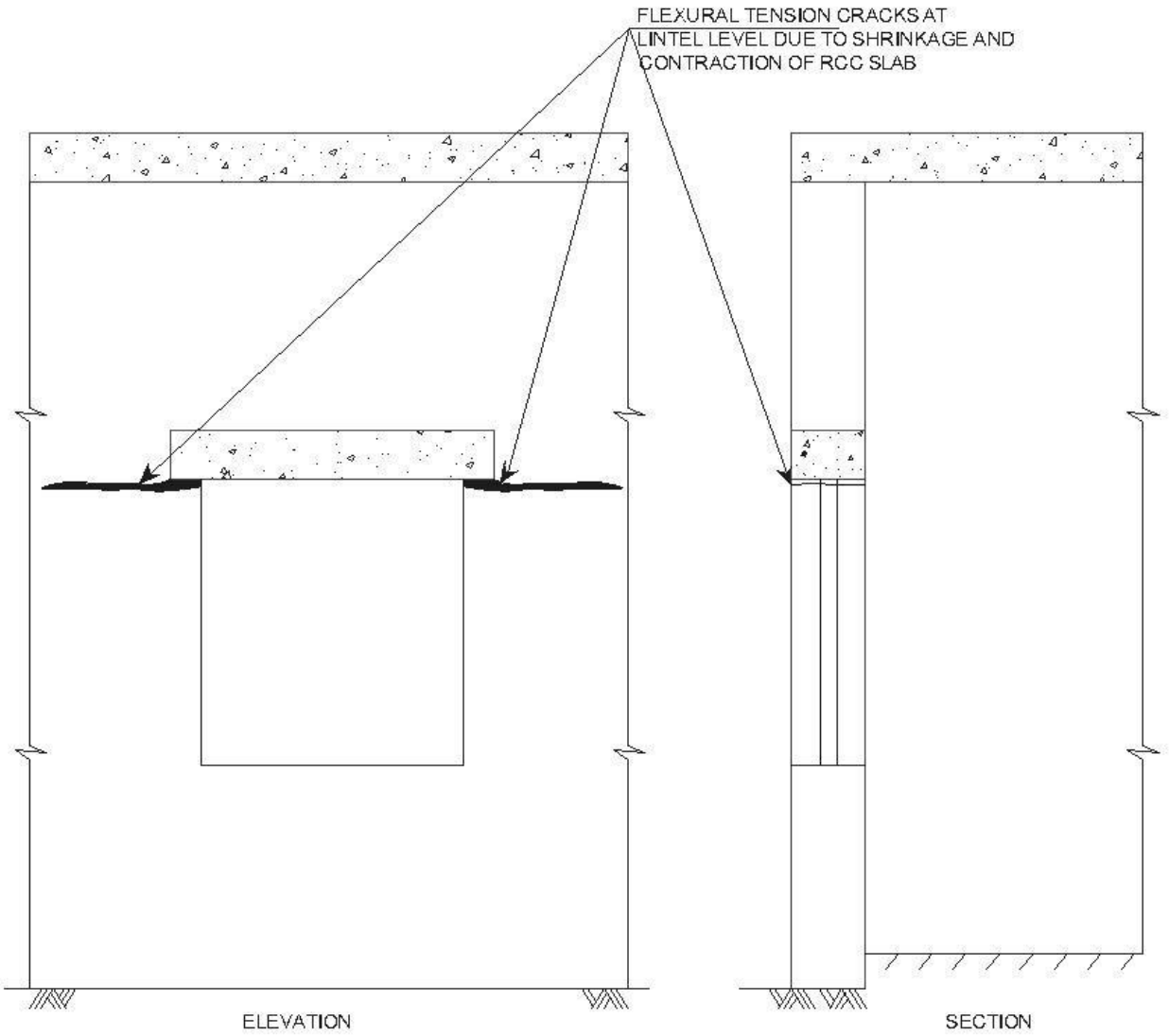


Tensile Crack in Masonry Wall

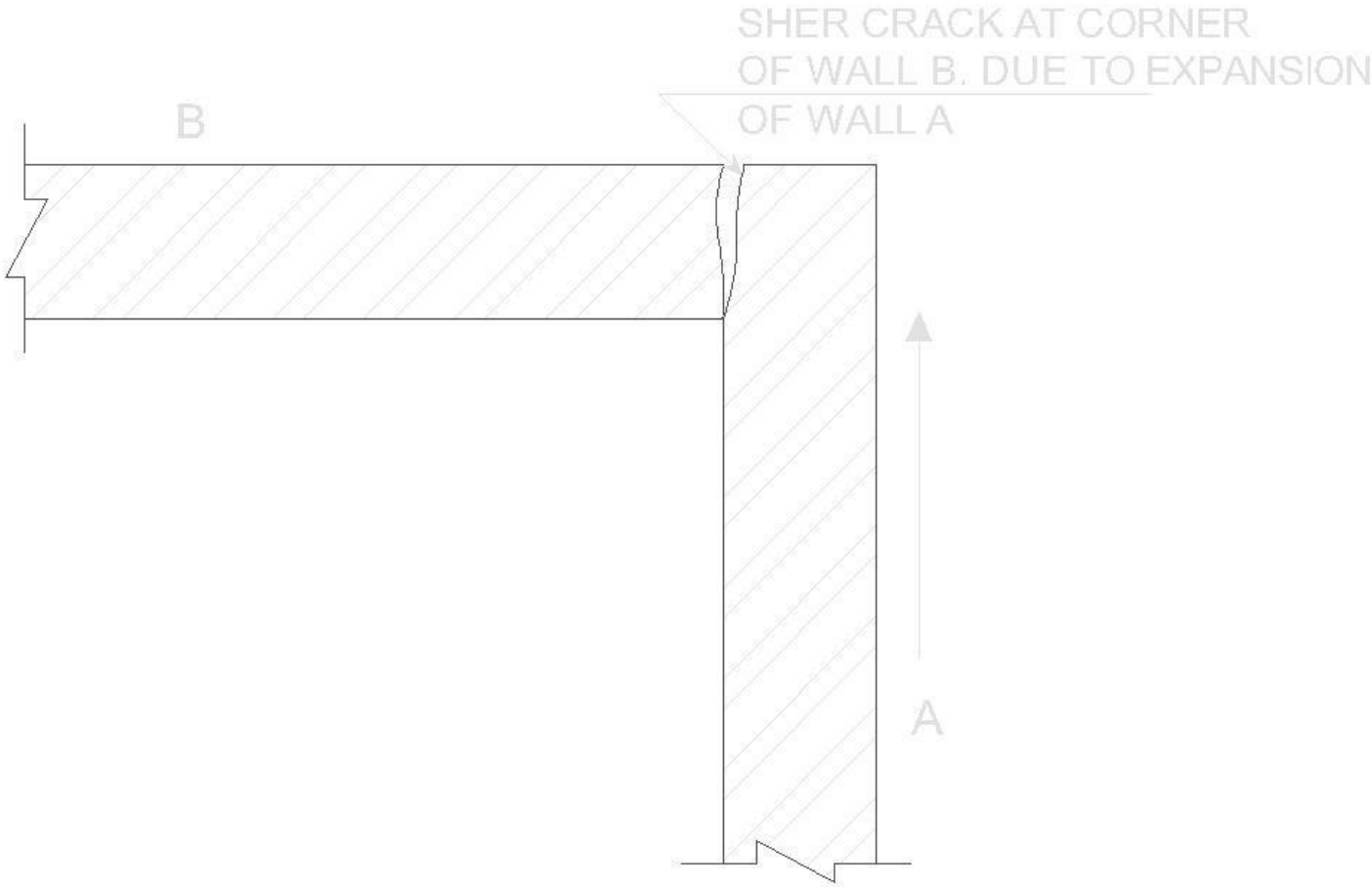
Present trend in construction

- Modern Structure
- Tall, slender, thin wall
- Designed for higher stress
- Constructed at fast pace
- More crack prone

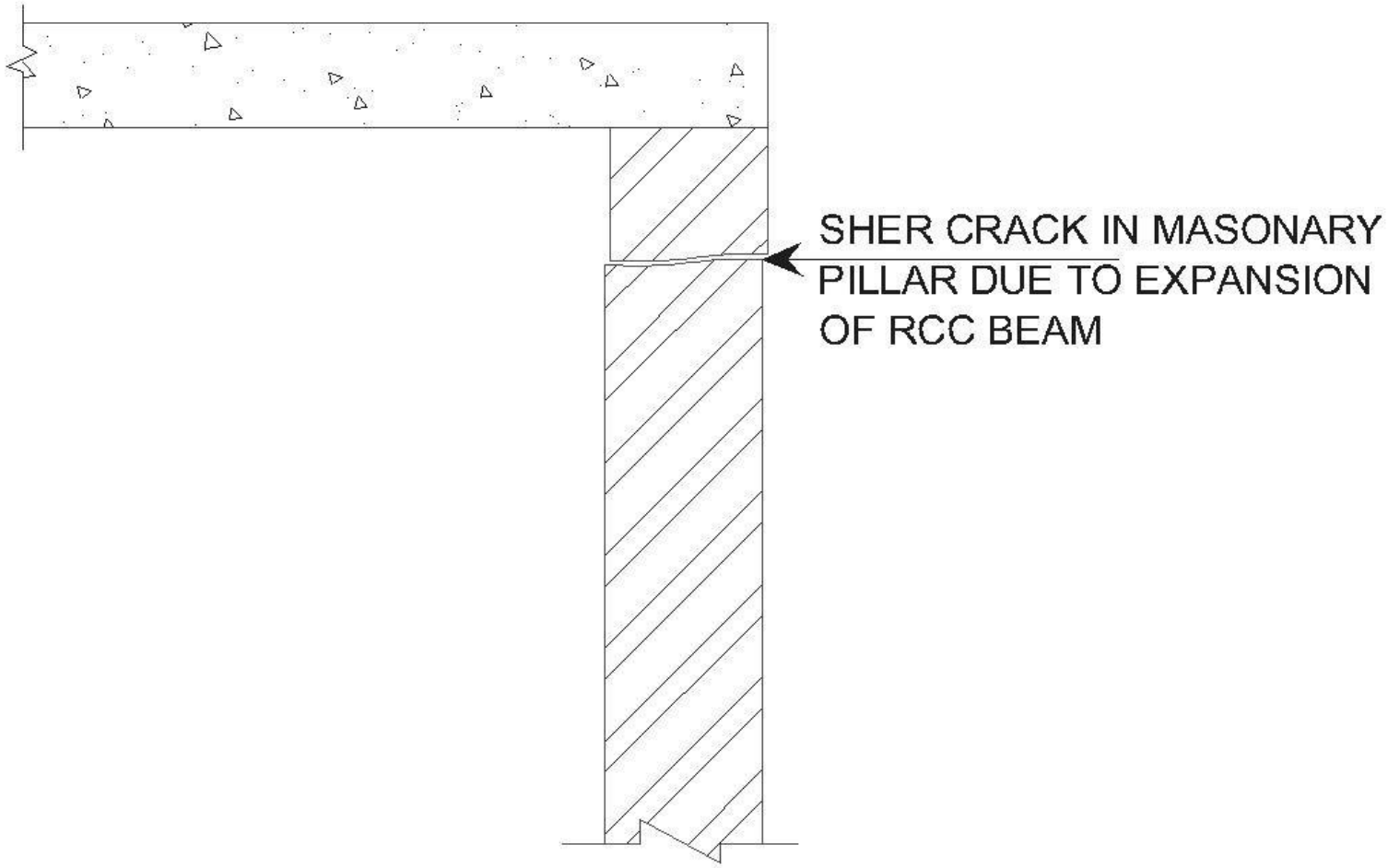




Flexural Tension Crack in Wall Masonry



Shear Crack in Masonary Wall



Shear Crack in Masonary Pillar
at Beam Support

Non structural cracks

- **Plastic shrinkage cracks**
- Cracking caused by plastic shrinkage in concrete occurs most commonly on the exposed surfaces of freshly placed floors and slabs with large surface areas when they are subjected to a very rapid loss of moisture caused by low humidity and wind or high temperature or both.
- Plastic shrinkage usually occurs prior to final finishing, before curing starts. When moisture evaporates from the surface of freshly placed concrete faster than it is placed by curing water, the surface concrete shrinks.
- In most cases, these cracks are wide at the surface. They range from a few millimeters to many meters in length and are spaced from a few centimeters to as much as 3 m apart.

Hairline cracks

- cracks may develop in concrete foundations as the concrete cures iscalled hairline cracks.
- If the cracks appear shortly after pouring the concrete foundation, concrete may have been mixed poorly or poured too quickly.
- In poured concrete foundations, hairline crack frequently appear in the center of the walls because the wall corners have a greater stability.

Settlement cracks

- They appear when the underlying ground has not been compacted or prepared properly or if the subsoil was not of the proper consistency.
- Settlement cracks are usually wider at the top of the crack than the bottom as the foundation "bends" over a single point, allowing differential settlement.

Thermal Cracking of Concrete

- Temperature difference within a concrete structure may be caused by portions of the structure losing heat of hydration at different rates or by the weather conditions cooling or heating one portion of the structure to a different degree or at a different rate than another portion of the structure.

- These temperature differences result in differential volume change, leading to cracks. This is normally associated with mass concrete including large and thicker sections of column, beams and slabs

THERMAL EXPANSION AND SHRINKAGE

- Expansion and contraction of concrete subjected to ambient increase or decrease in temperature results concrete cracking.
- Ex: roof slabs, road or airfield pavements , bridge decks etc.

Cause of cracks in buildings

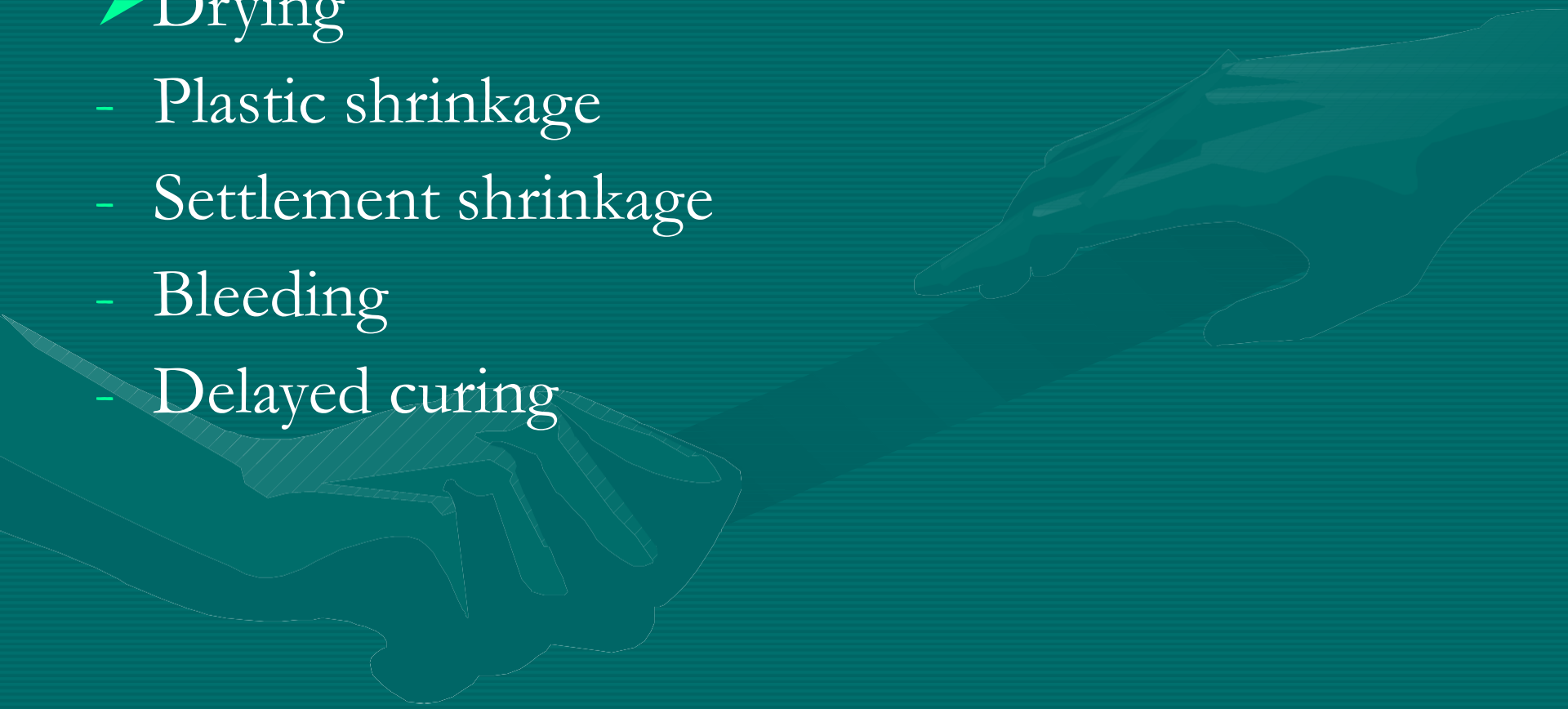
- Moisture change
- Thermal variation
- Elastic deformation
- Creep
- Chemical reaction
- Foundation movement and settlement of soil
- Vegetation

Moisture Movement

- Reversible Movement
- Material expands on absorbing moisture content
- Shrinks on drying
- Irreversible movement
- Material undergo some irreversible movement due to initial moisture change

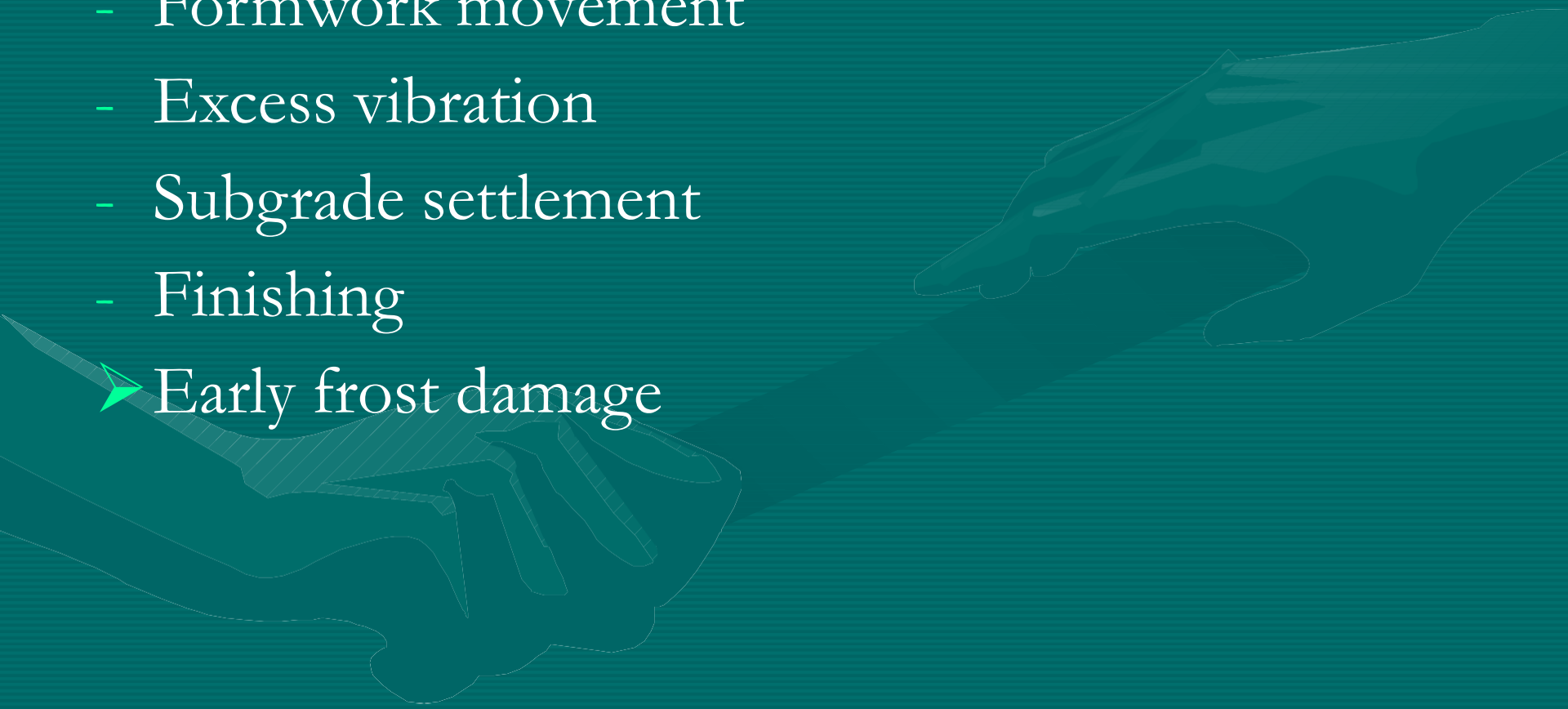
Types and causes of cracks in Concrete

- Before hardening
  - Drying
  - Plastic shrinkage
  - Settlement shrinkage
  - Bleeding
  - Delayed curing



Continue…

- Constructional
  - Formwork movement
  - Excess vibration
  - Subgrade settlement
  - Finishing
- Early frost damage



Continue…

- After hardening
  - Unsound material
  - Long term drying shrinkage
  - Thermal
  - Moisture movement
  - Biological
  - Structural design deficiencies
  - Chemical
  - Corrosion of reinforcement

Plastic shrinkage cracks

- Concrete surface loses water faster than the bleeding action brings it to top
- Quick drying of concrete at the surface results in shrinkage
- Concrete at the plastic state can not resist tension
- Crack Depth 5 to 10 cm, width 3 mm
- Once developed difficult to rectify

Measure to reduce plastic crack

- Moisture the sub grade and form work
- Erect the temporary wind breaker
- Erect the temporary roof to protect green concrete from hot sun
- Reduce the time between placing & finishing
- In case of delay in finishing cover the concrete with polythene

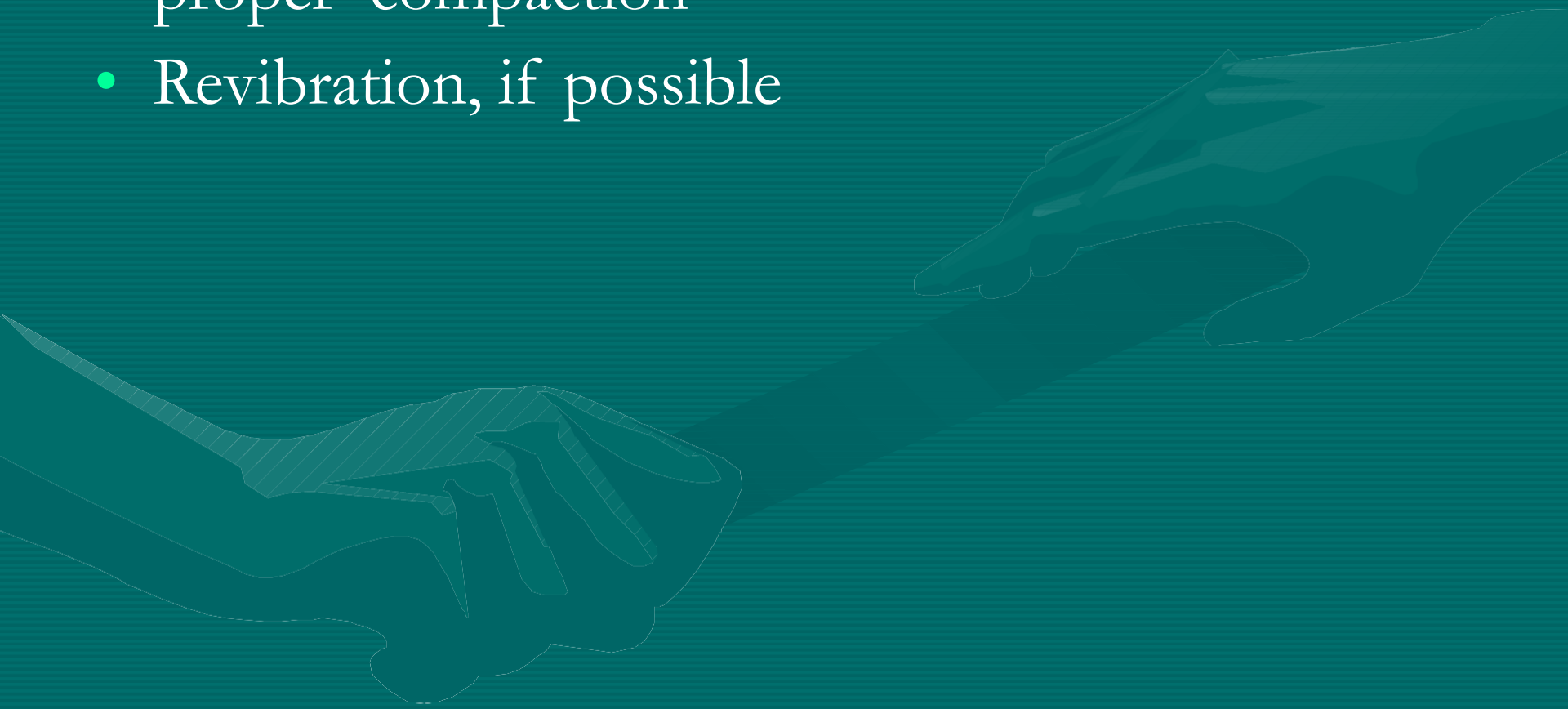
Settlement shrinkage

- If concrete is free to settle uniformly, no crack occurs
- Obstruction to uniform settlement creates voids/cracks (Reinforcement/Aggregate)
- Settlement crack (common in deep beam)



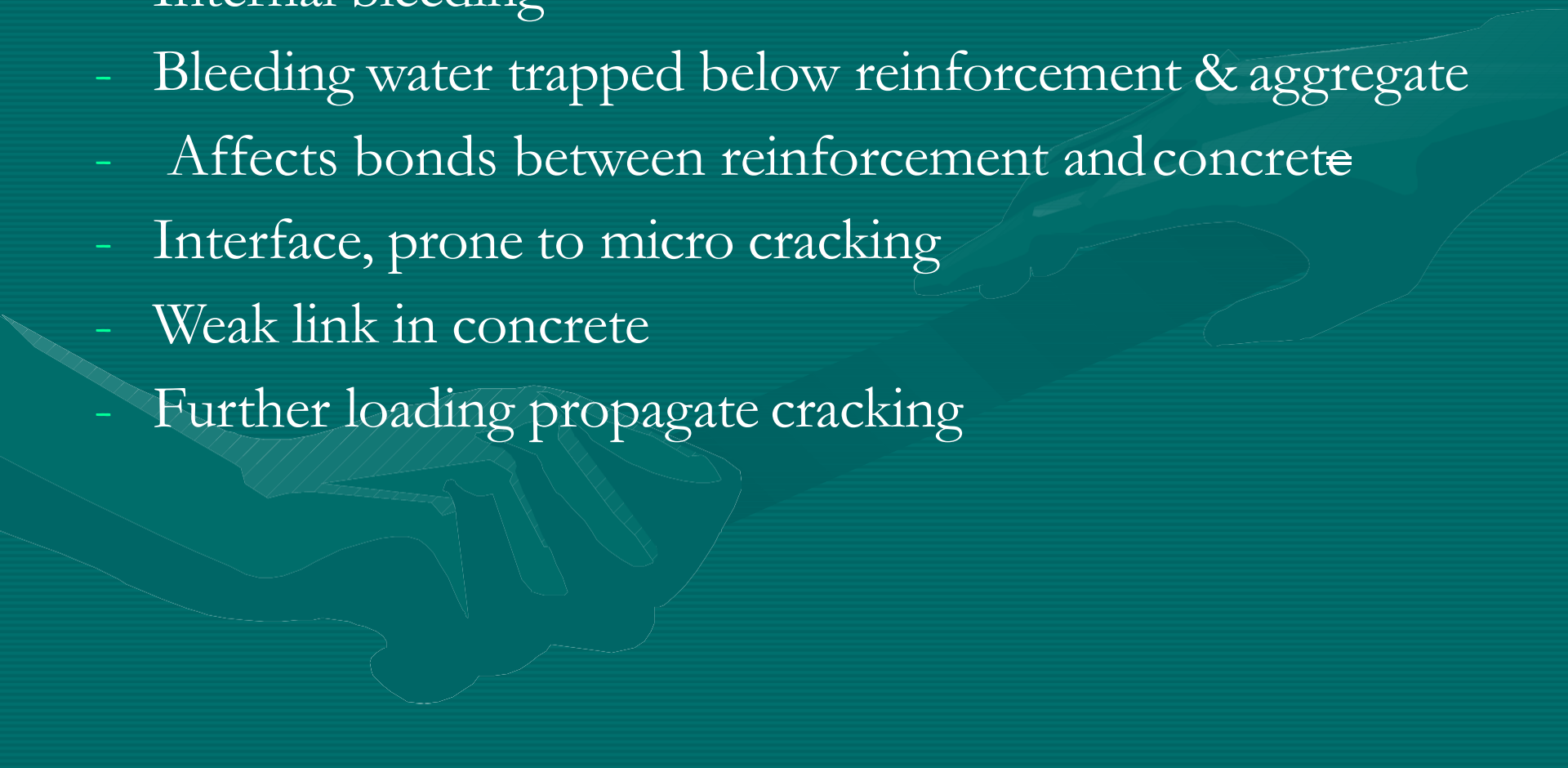
Measure

- Pouring of concrete in layers with proper compaction
- Revibration, if possible



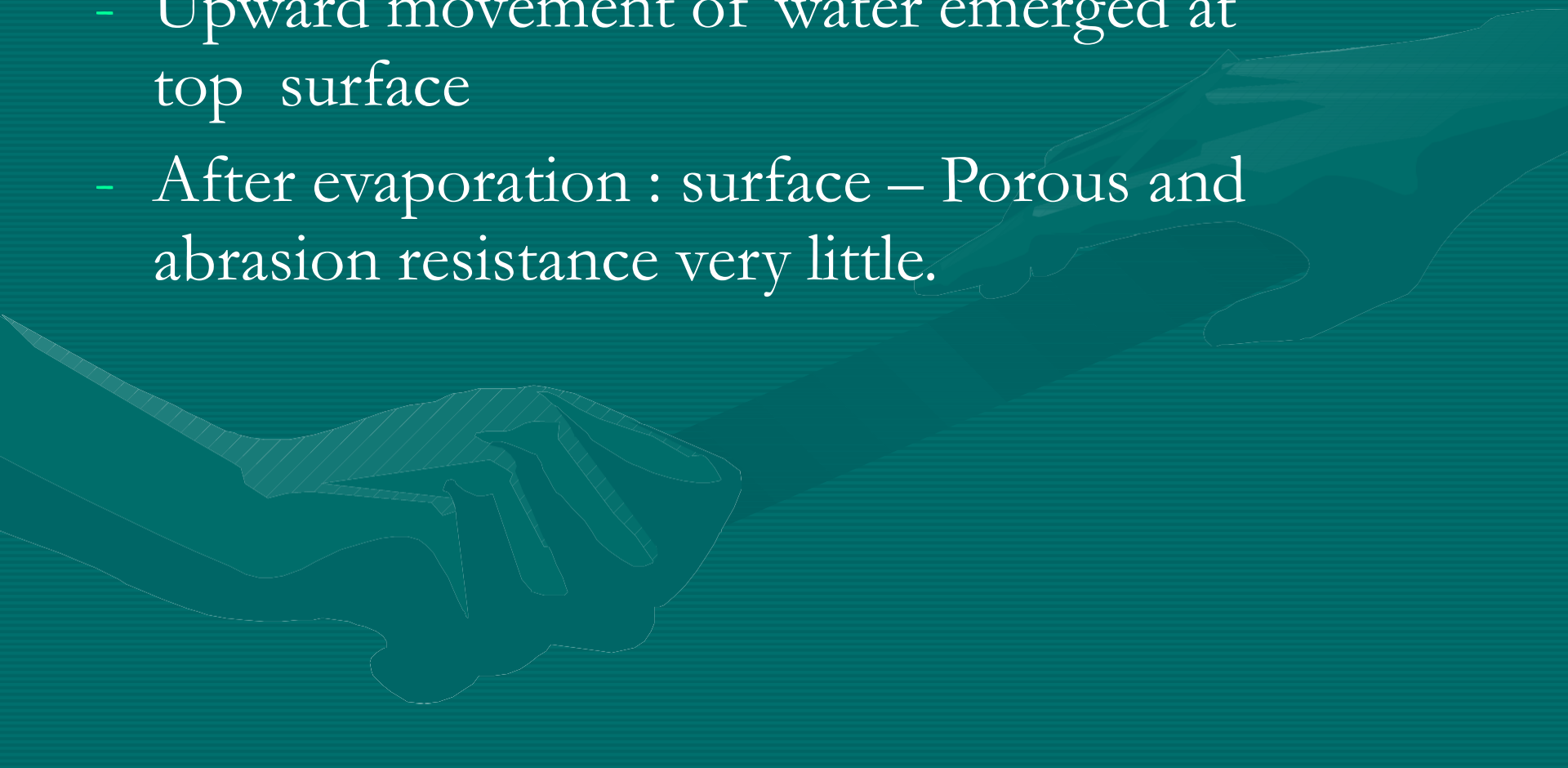
Bleeding

- Upward movement of water when concrete settle downs
- Internal bleeding
- Bleeding water trapped below reinforcement & aggregate
- Affects bonds between reinforcement and concrete
- Interface, prone to micro cracking
- Weak link in concrete
- Further loading propagate cracking



Continue…

- External Bleeding
- Upward movement of water emerged at top surface
- After evaporation : surface – Porous and abrasion resistance very little.



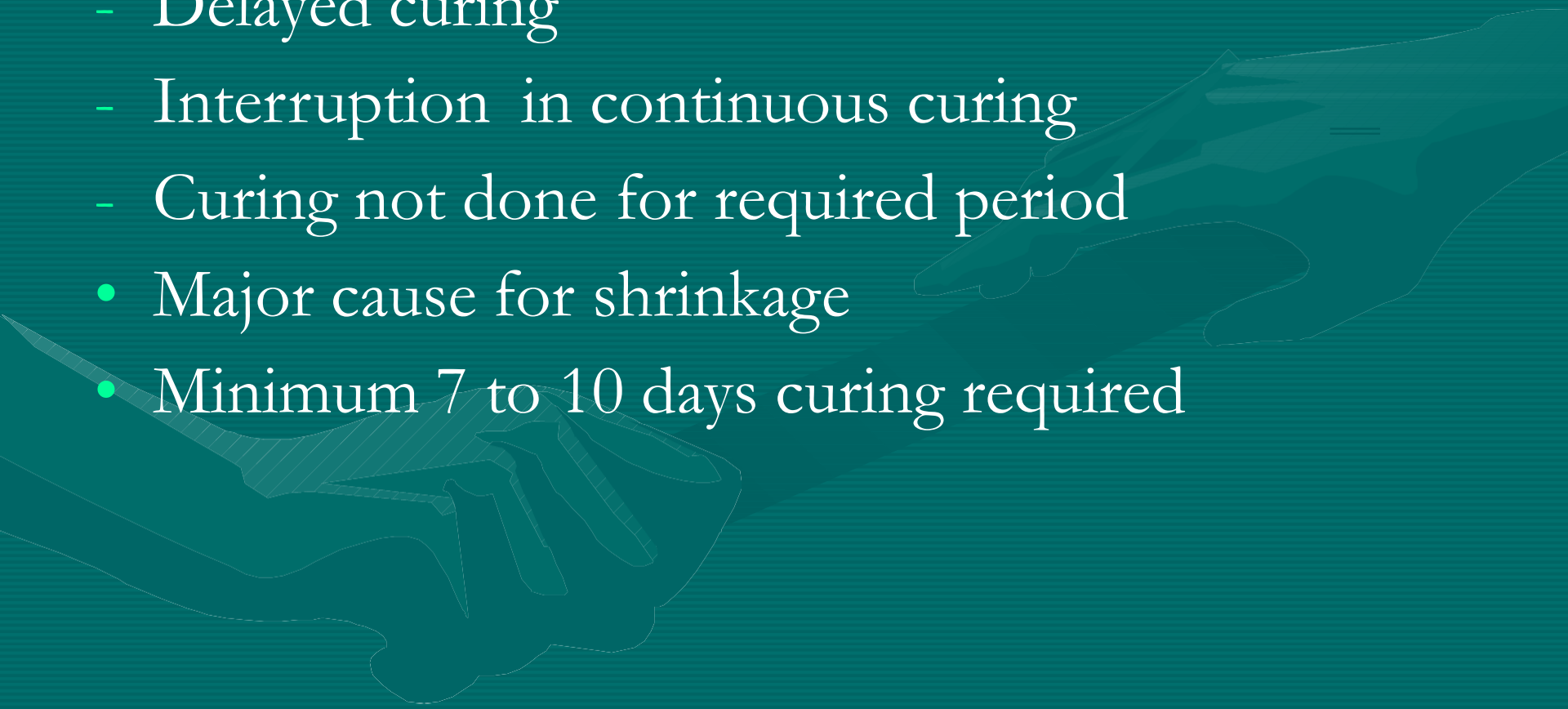
General Observations

- Mason floats concrete when bleeding water still standing.
- Results
- Downward movement of coarse aggregate
- Upward movement of fine particles (cement and water)
- Top surface
- Presence of fine materials develops crack and craziness

(Craziness: occurrence of closely spaced crack at surface)

Delayed Curing

- Common practice
- Delayed curing
- Interruption in continuous curing
- Curing not done for required period
- Major cause for shrinkage
- Minimum 7 to 10 days curing required



Constructional Effects

- Crack/Deformation of plastic concrete (After Compaction)

-Lack of rigidity of formwork (Remains unnoticed)

-Use of high consistency concrete (present trend)

: Pumping requirement, Use of superplasticizer

-Avoid segregation by proper vibration

- Segregated Concrete Mix- Exhibits high shrinkage crack

Unsound Material

- Use of crushed sand very common now
- Amount of dust (<75 micron): Very high
- Presence of excess dust :
  - Causes crack formation
  - Interfere with setting time, Shrinkage and Bond
- Maximum %age of fine = 3% (Coarse aggregate)

CRACK REPAIR TECHNIQUES

1. Sealing with epoxies
2. Routing and sealing
3. Stitching
4. External stressing
5. Overlays
6. Grouting
7. Blanketing
8. Autogenous healing

1.SEALING WITH EPOXIES

- Injecting epoxy bonding compounds in high pressure in to cracks

PROCEDURE

1. Drill into the cracks
2. Flush out cracks by injecting water/other solvents
3. Dry the surface
4. Epoxy injection in to holes
5. Curing of epoxy
6. Remove surface seal by grinding





2. ROUTING AND SEALING

- Simplest , most common , inexpensive method
- For both fine and larger isolated cracks
- This method involves enlarging the crack along its exposed face and sealing it with a suitable joint sealant
- Most used for floors and pavements
- In road pavements hot tar used as sealant
- Side effects –
  - chemical attack
  - corrosion of rebars
  - swelling

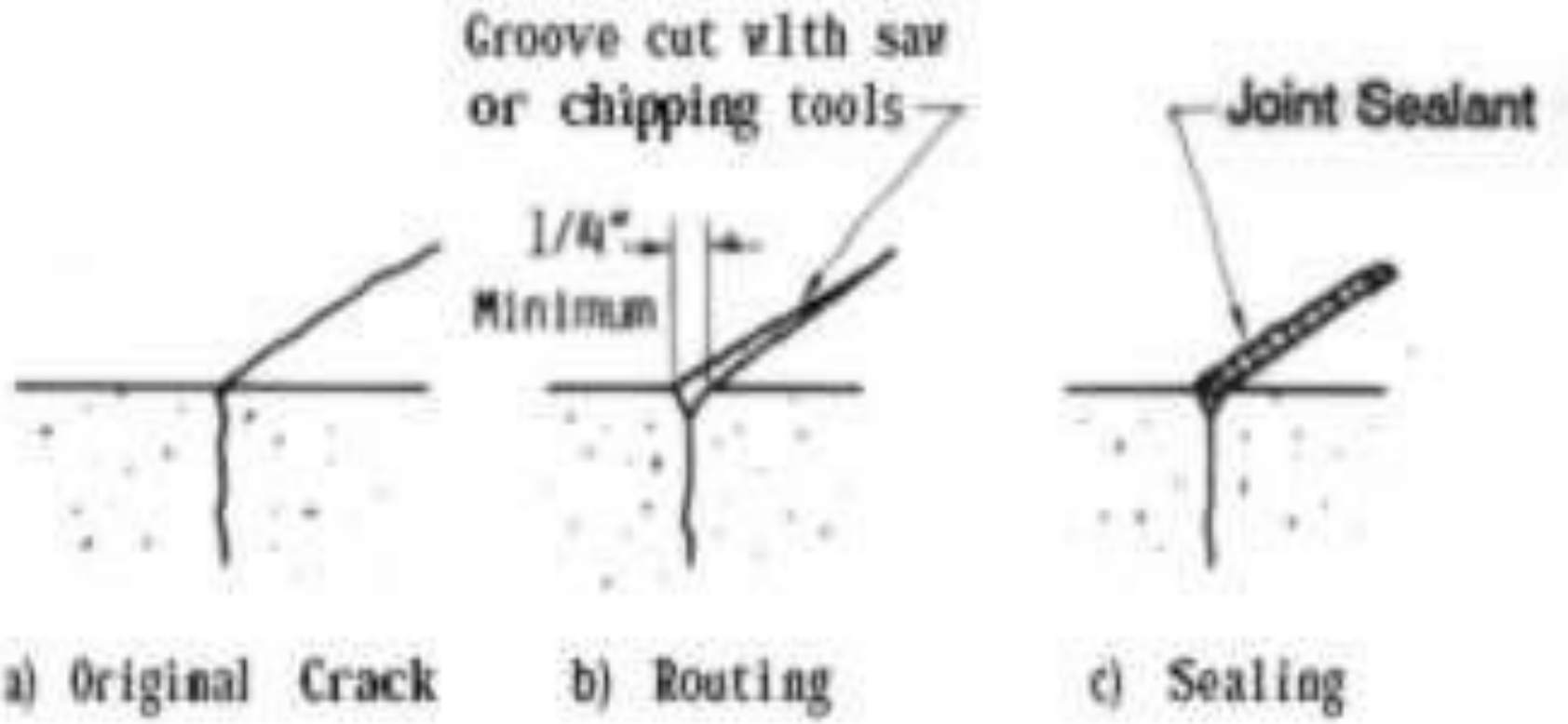
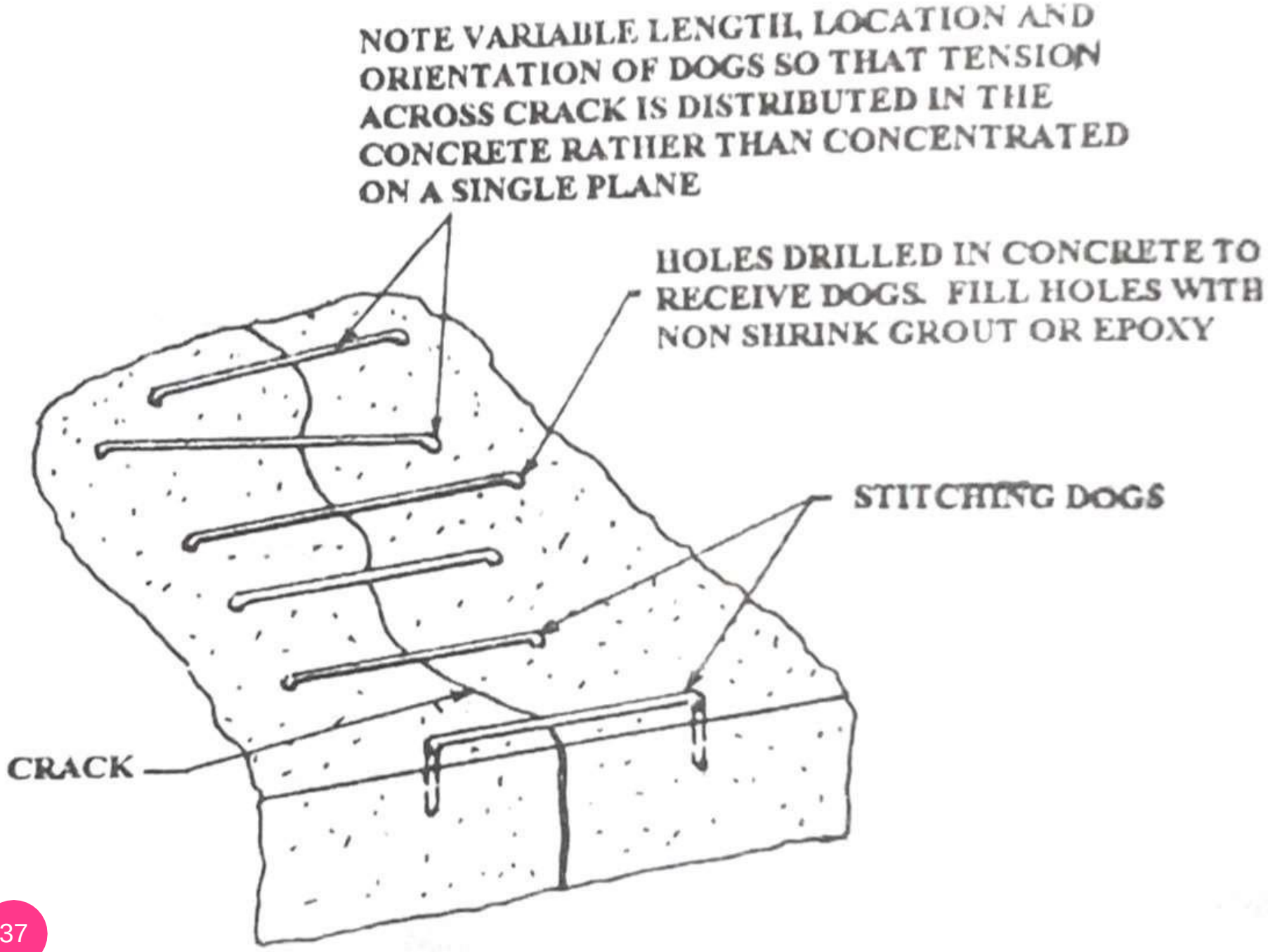


Fig. 3.1-Repair of crack by muting and sealing (Johnson 1965)

3. STITCHING

- Stitching may be used when tensile strength must be reestablished across major cracks.
- Stitching involves
  - drilling holes on both sides of the crack
  - grouting in U-shaped metal units with short legs called staples or stitching dogs


NOTE VARIABLE LENGTH, LOCATION AND ORIENTATION OF DOGS SO THAT TENSION ACROSS CRACK IS DISTRIBUTED IN THE CONCRETE RATHER THAN CONCENTRATED ON A SINGLE PLANE
HOLES DRILLED IN CONCRETE TO RECEIVE DOGS. FILL HOLES WITH NON SHRINK GROUT OR EPOXY
STITCHING DOGS
CRACK



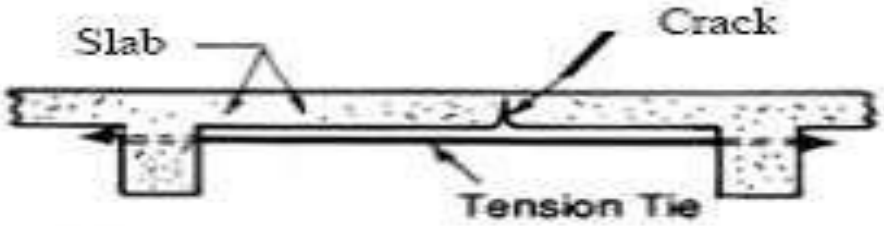


4. EXTERNAL STRESS

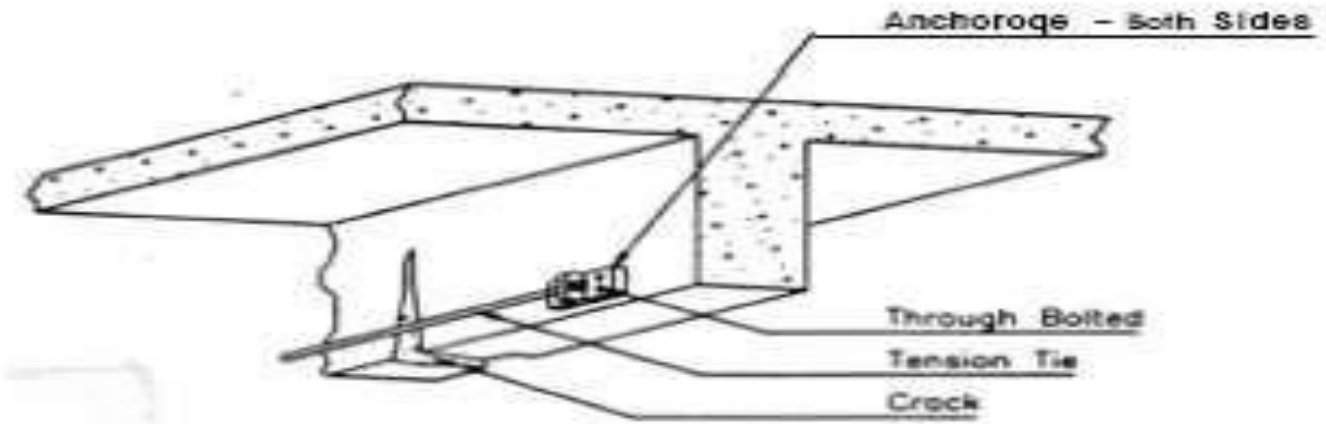
- The development of cracking is due to the tensile stress ,thus can be arrested by suppressing this stress
- Cracks can be closed by inducing a compression force to over come the tensile stresses
- The compressive force is applied by

1. Pre-stressing wires or rods
2. Wedging – by opening the cracks and filling with expanding mortar,by jacking and grouting or by actual driving wedges

Examples of external stressing



a) To Correct Cracking of Slab



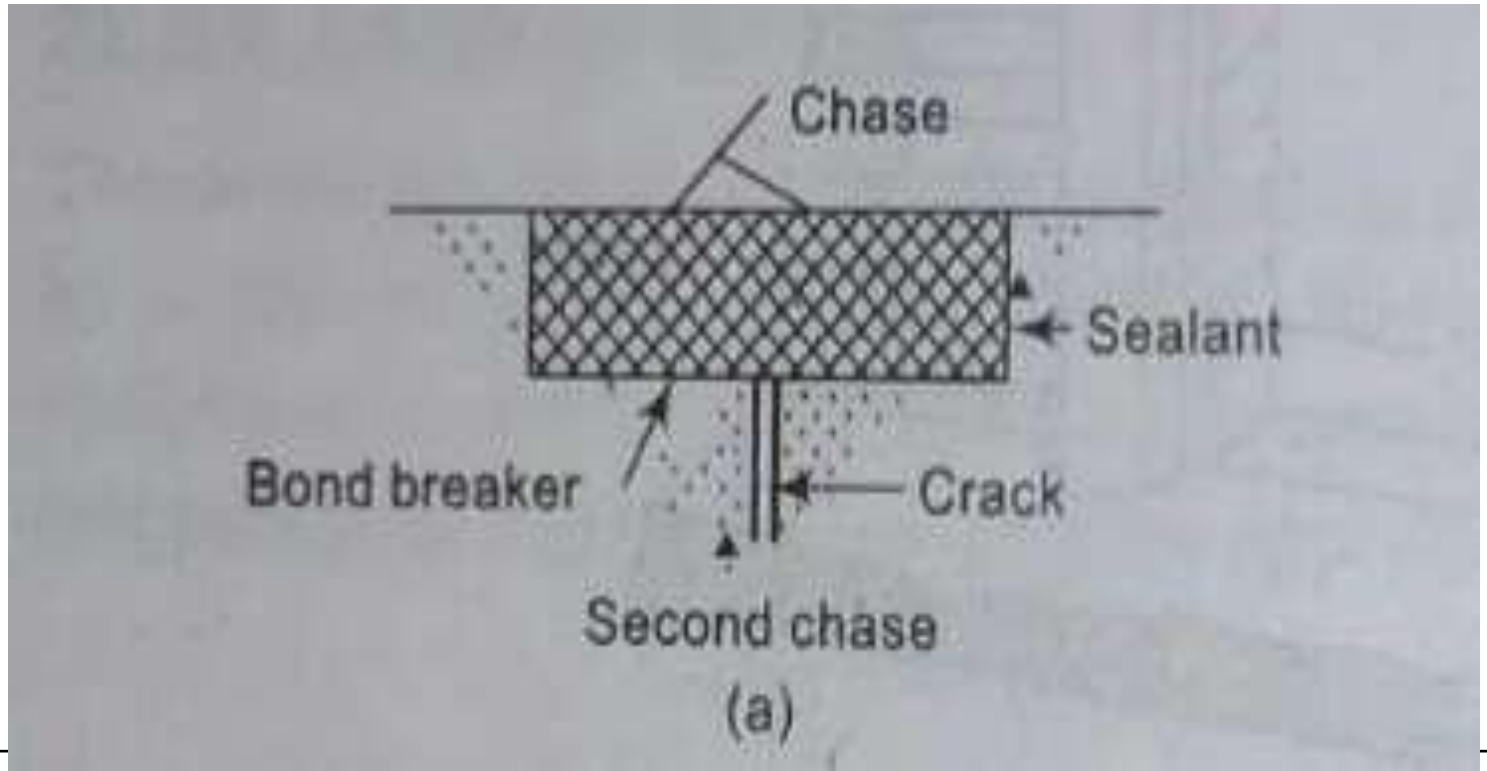
b) To Correct Cracking of Beam

Fig 3.5-Examples of external prestressing (Johnson 1965)

5.BLANKETING

- Blanketing is similar to routing and sealing
- used on a larger scale and is applicable for sealing active as well as dormant cracks.
- Following are the types of blanketing joints
- Type I
- Type II
- TypeIII
- Type IV

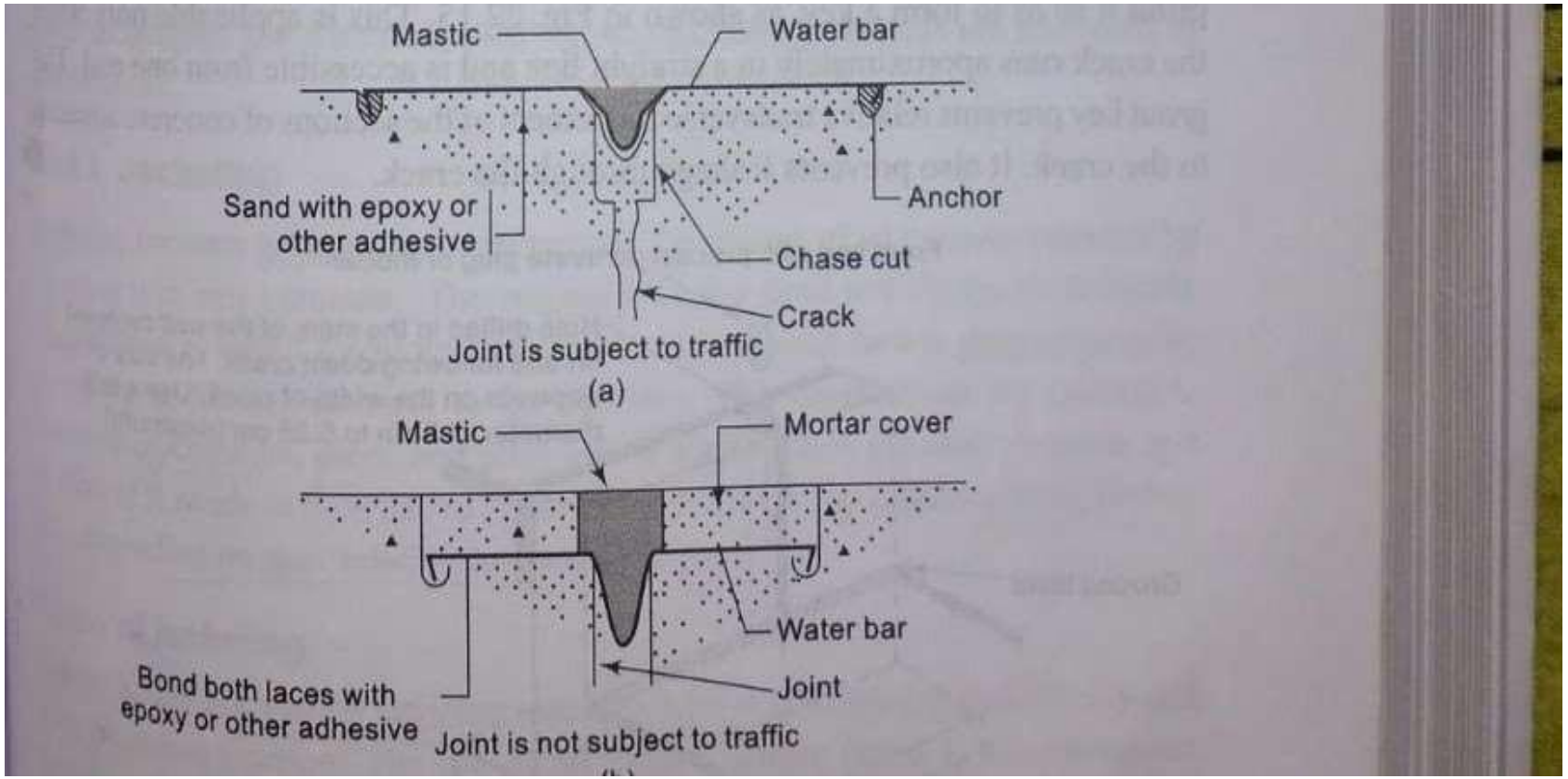
- Type I
- The first type of blanket joints use elastic sealants
- They return to their original shape, when not under an externally induced stress
- A bond breaker should be used at the bottom of the chase, so that the sealant is free to deform.



- Type II
- use sealant materials that are known as mastic sealants
- Their details are similar to that of an elastic sealant, except that the bond breaker is omitted and the sealant is bonded to the bottom as well as to the sides of the chase.

- Type III
- It is a mortar plugged joint
- A recess in the form of a trapezoid to accomodate the mortar plug is made
- This recess is filled with mortar

- Type IV
- A water cripped bar is used



6 .OVERLAYS

- Used to seal cracks
- Used when large no of cracks, treating each crack is expensive
- Active cracks – overlays done with materials which are extensible but not flexible

  Eg : Polymeric membrane with top coat of tar
- Dormant cracks – any type of overlays may be used

  Eg: polymer modified Portland cement mortar or concrete, or by silica fume concrete


BEFORE
AFTER

7. GROUTING

- Similar to epoxy injection
- Epoxy not used where fire resistance and cold weather
- Grouting is effective alternative
- When the crack is straight line
- Drill out the length of crack
- grout it to form a key
- This method is effective in stopping water leaks

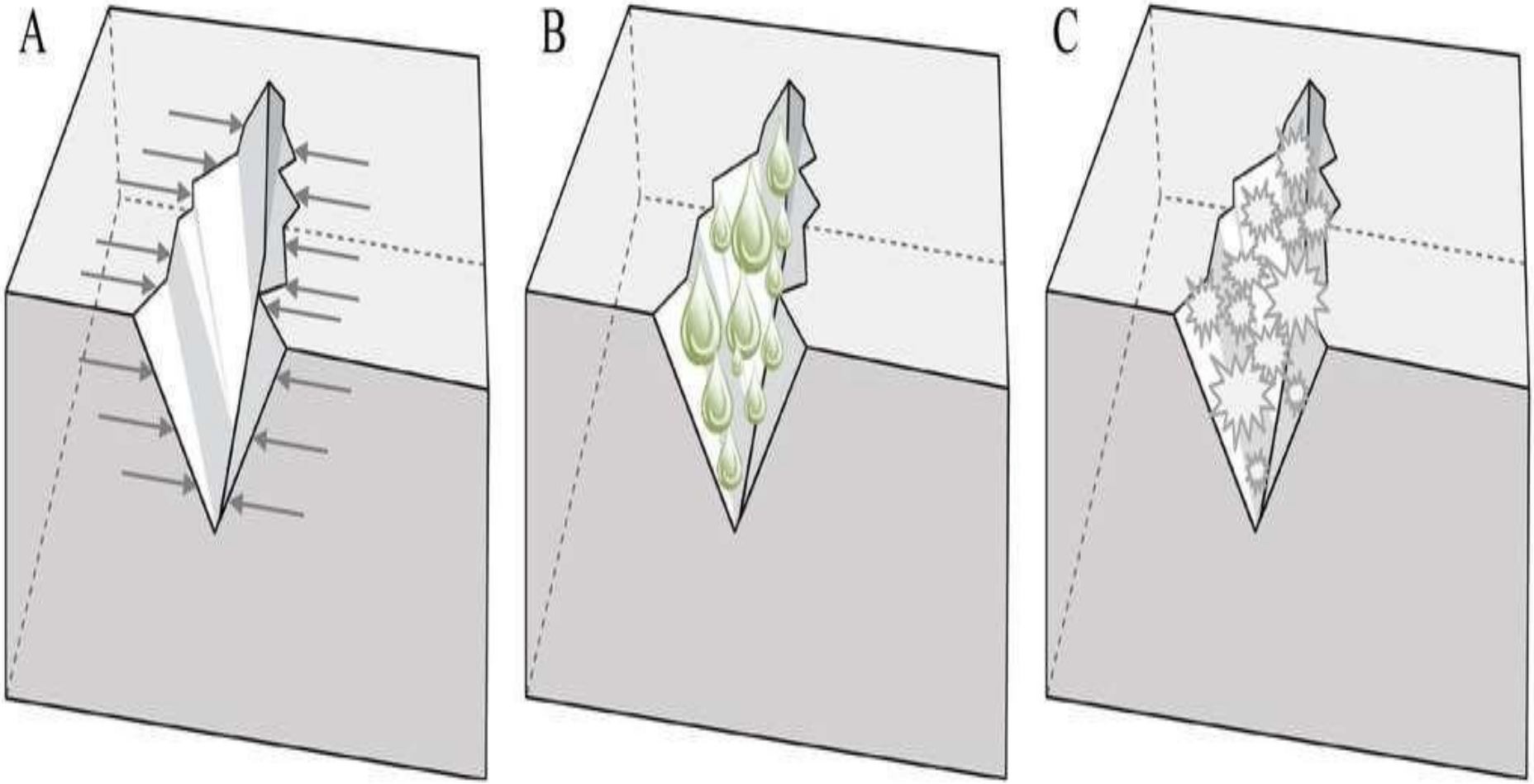
8.AUTOGENOUS HEALING

- Autogenous healing is the natural process of crack repair that can occur in concrete in the presence of moisture
- The repair is by a combination of mechanical blocking by particles carried into the crack with the water and the deposition of calcium carbonate from the cementitious material
- MECHANISM

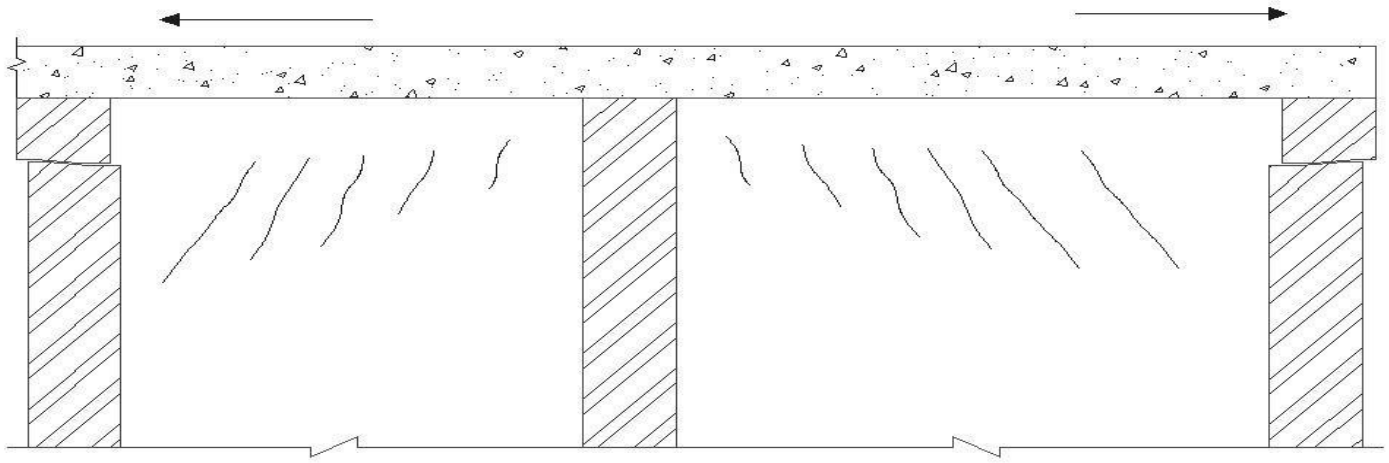
Autogenous healing occurs by the carbonation of calcium oxide and calcium hydroxide present in the cement by CO_2 present in the air and water

The resulting $CaCO_3$ and $Ca(OH)_2$ crystals precipitate accumulate and grow through and out from cracks

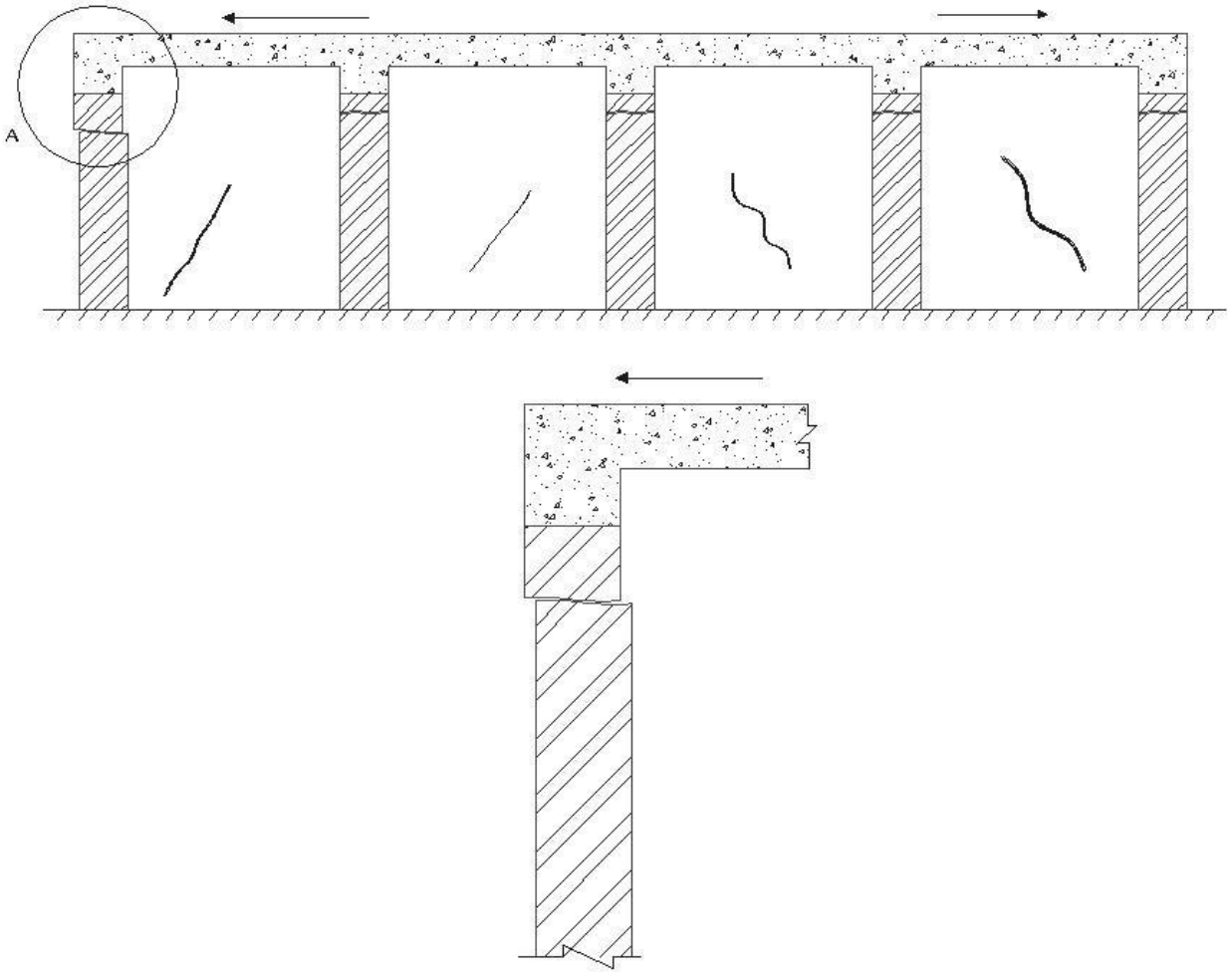
Develop a mechanical and chemical bonding between crystals and between surface and crystals


A
B
C

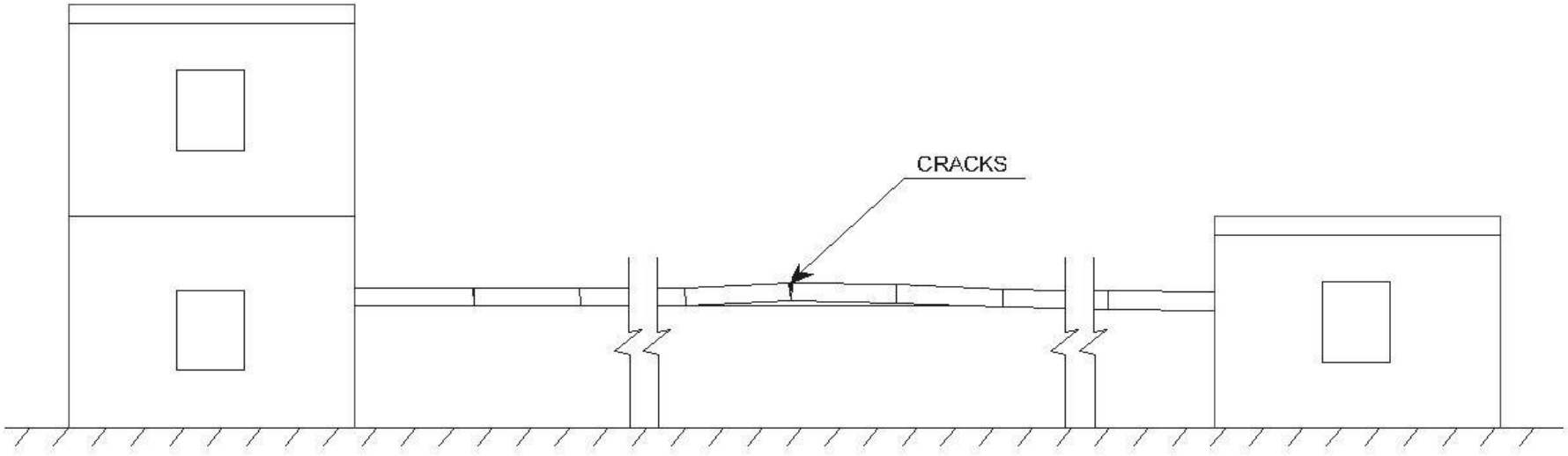
Some measures to avoid cracks



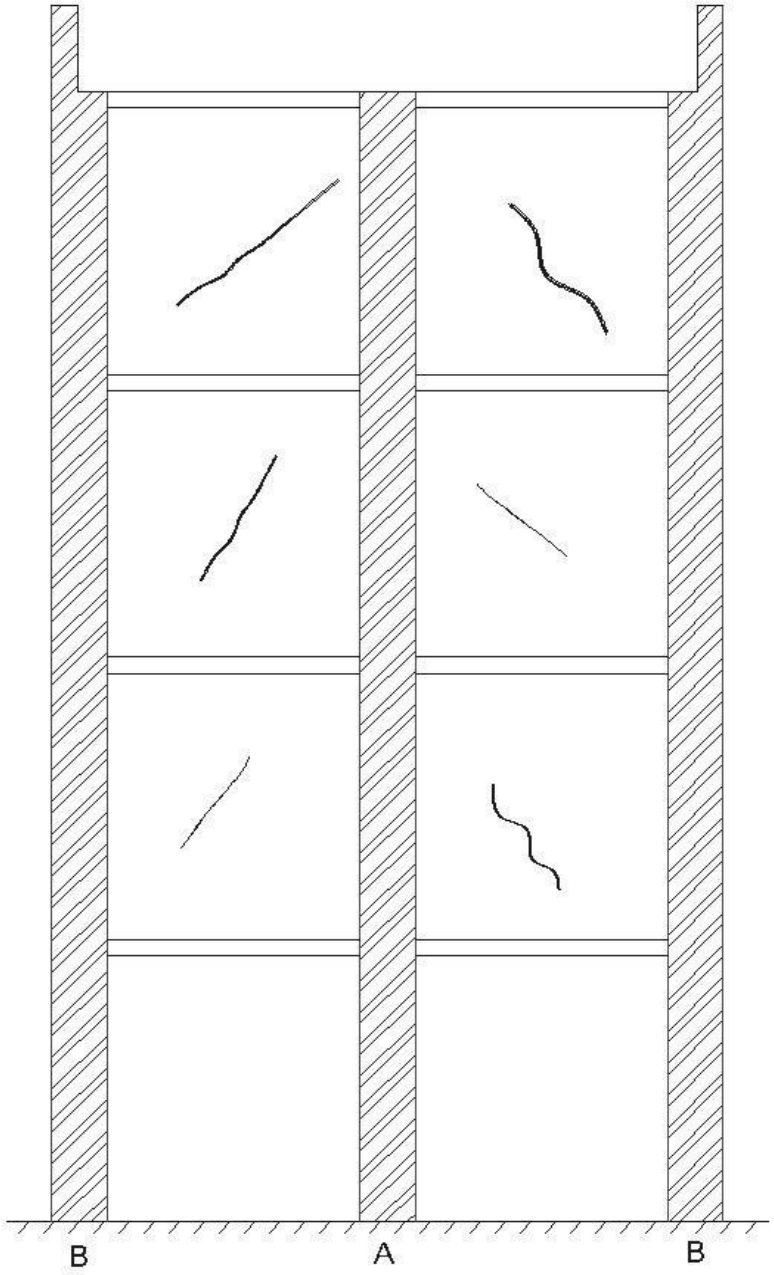
Cracking in Top Most Storey of a Load Bearing Structure



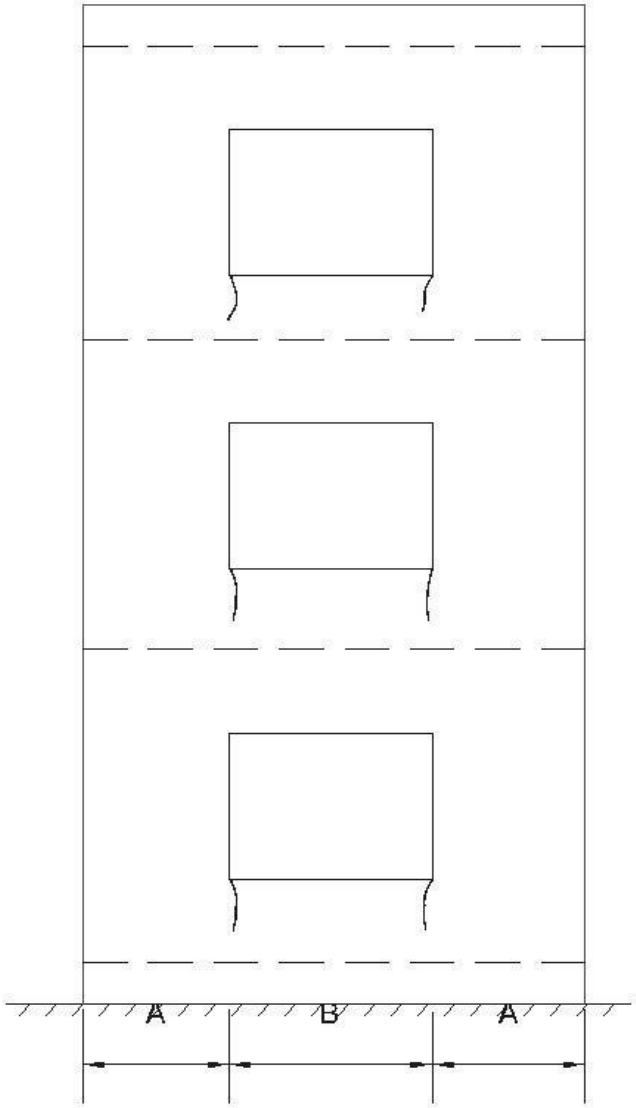
Cracking in Cladding and Cross Walls of a Framed Structure



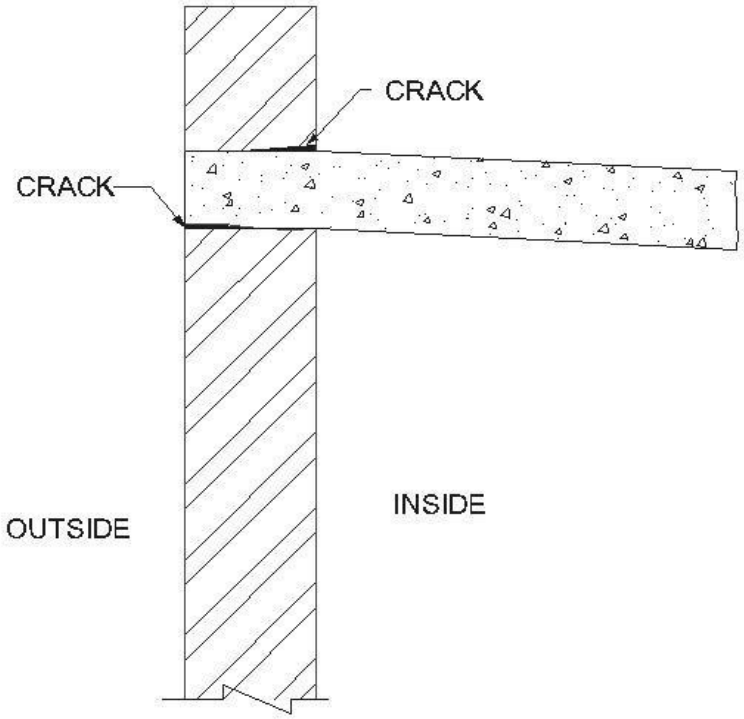
Arching up and Cracking of Coping of a Long Garden Wall



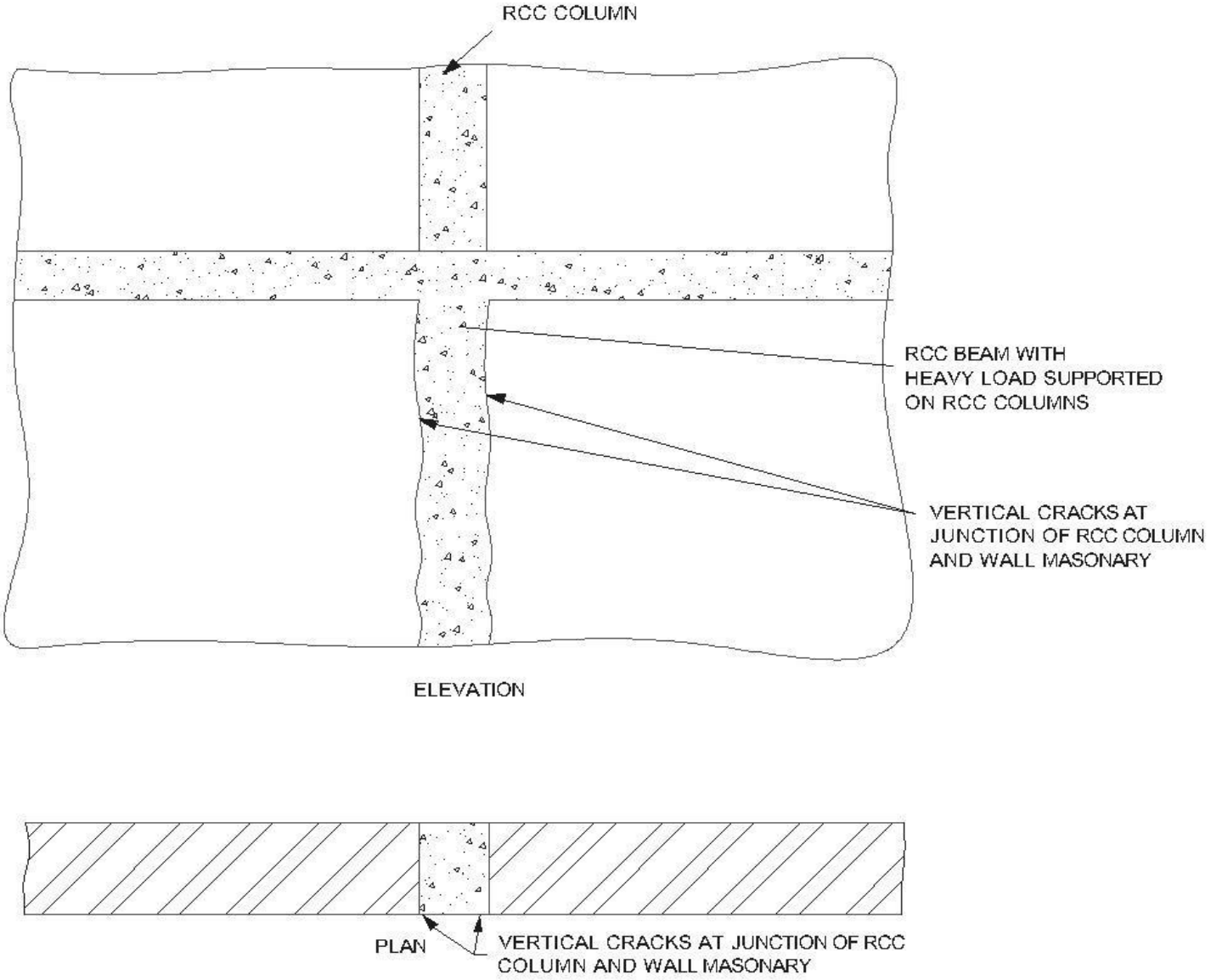
Diagonal Cracks in Cross Walls of Multi-Storied Load Bearing Structures



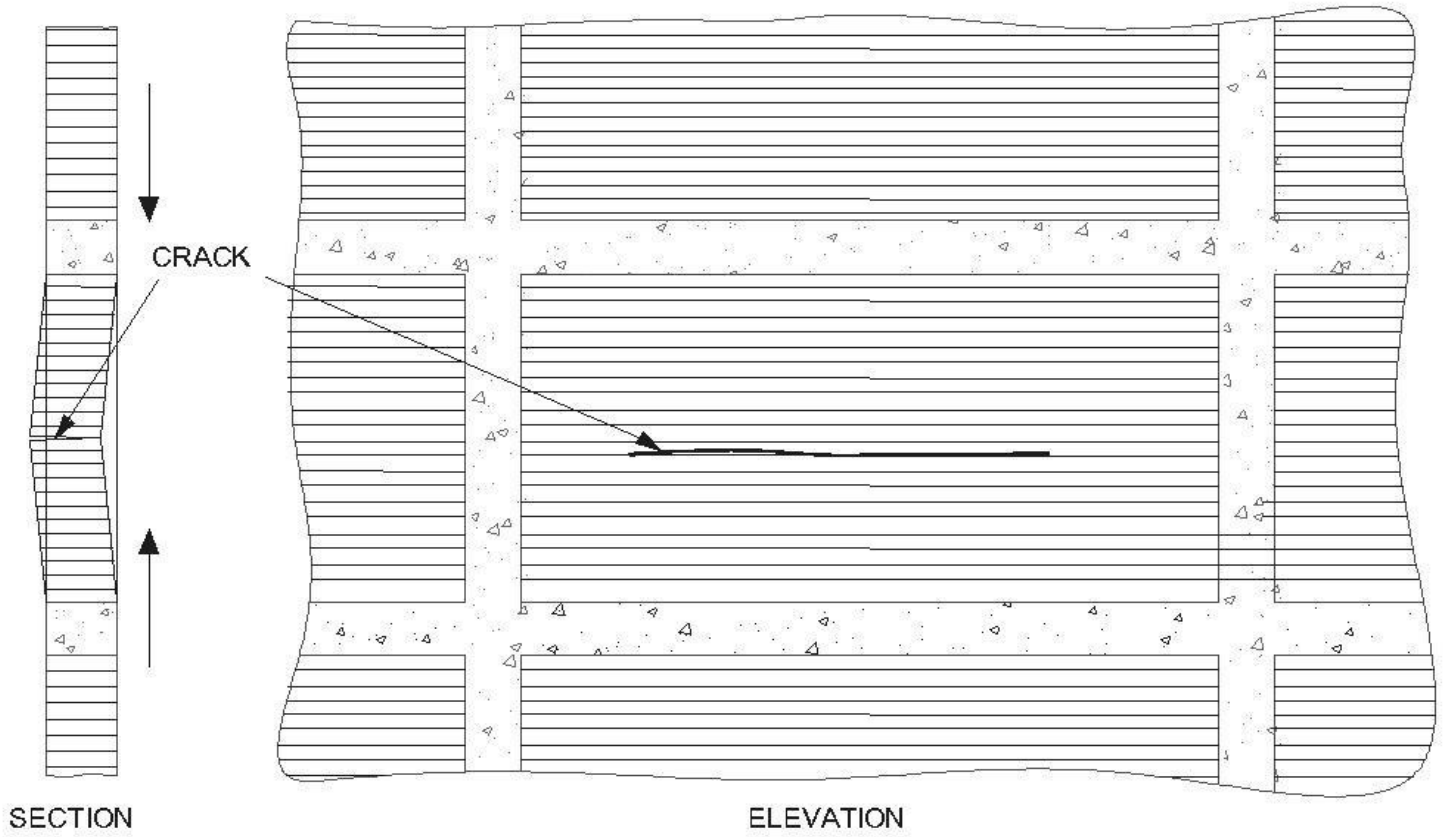
Vertical Cracks in Multi Storyed Building Having Window Openings in Load Bearing Wall



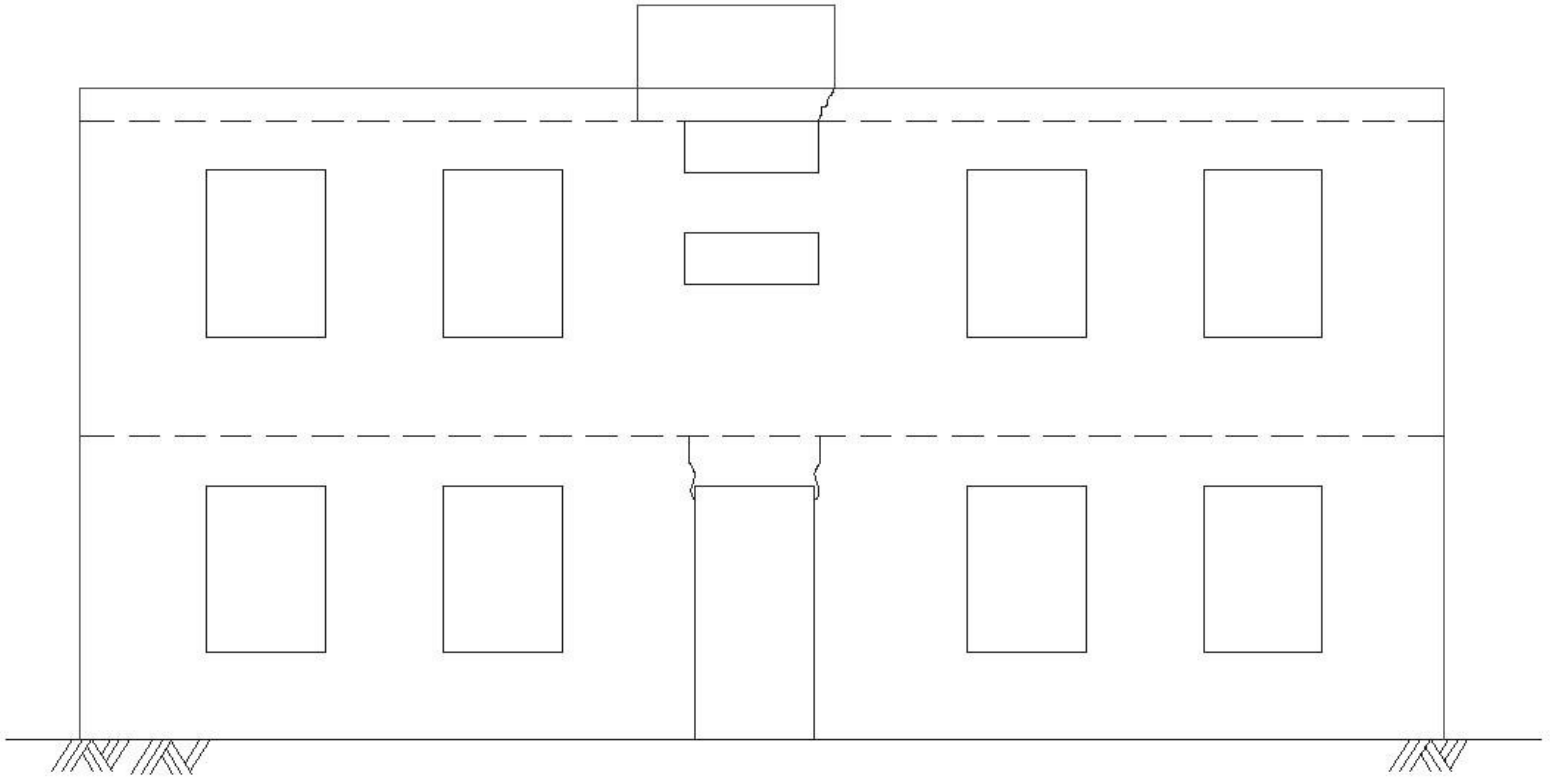
Horizontal Cracks in a Wall at Supports Due to Excessive Deflection of a Slab of Large Span



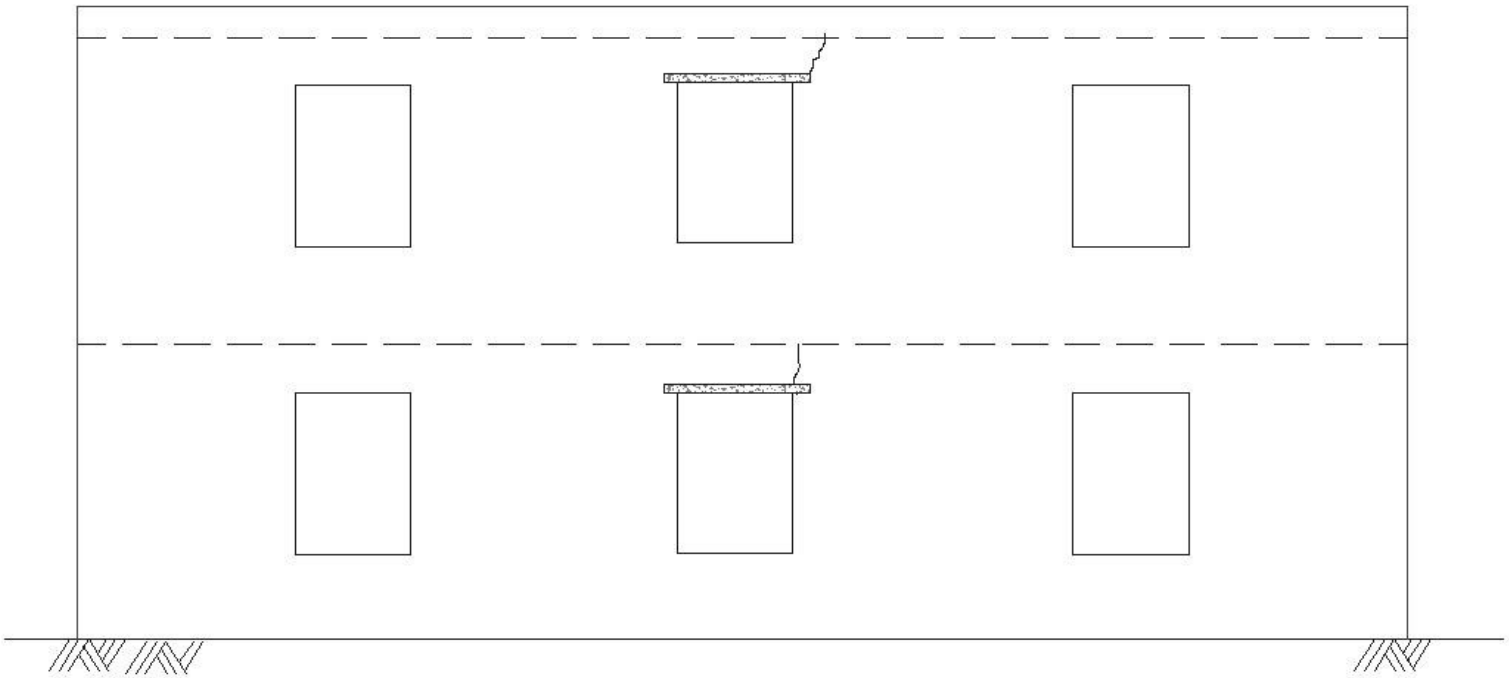
Vertical Cracks at Juction of RCC Column and Wall Masonary in a Load Bearing Structure



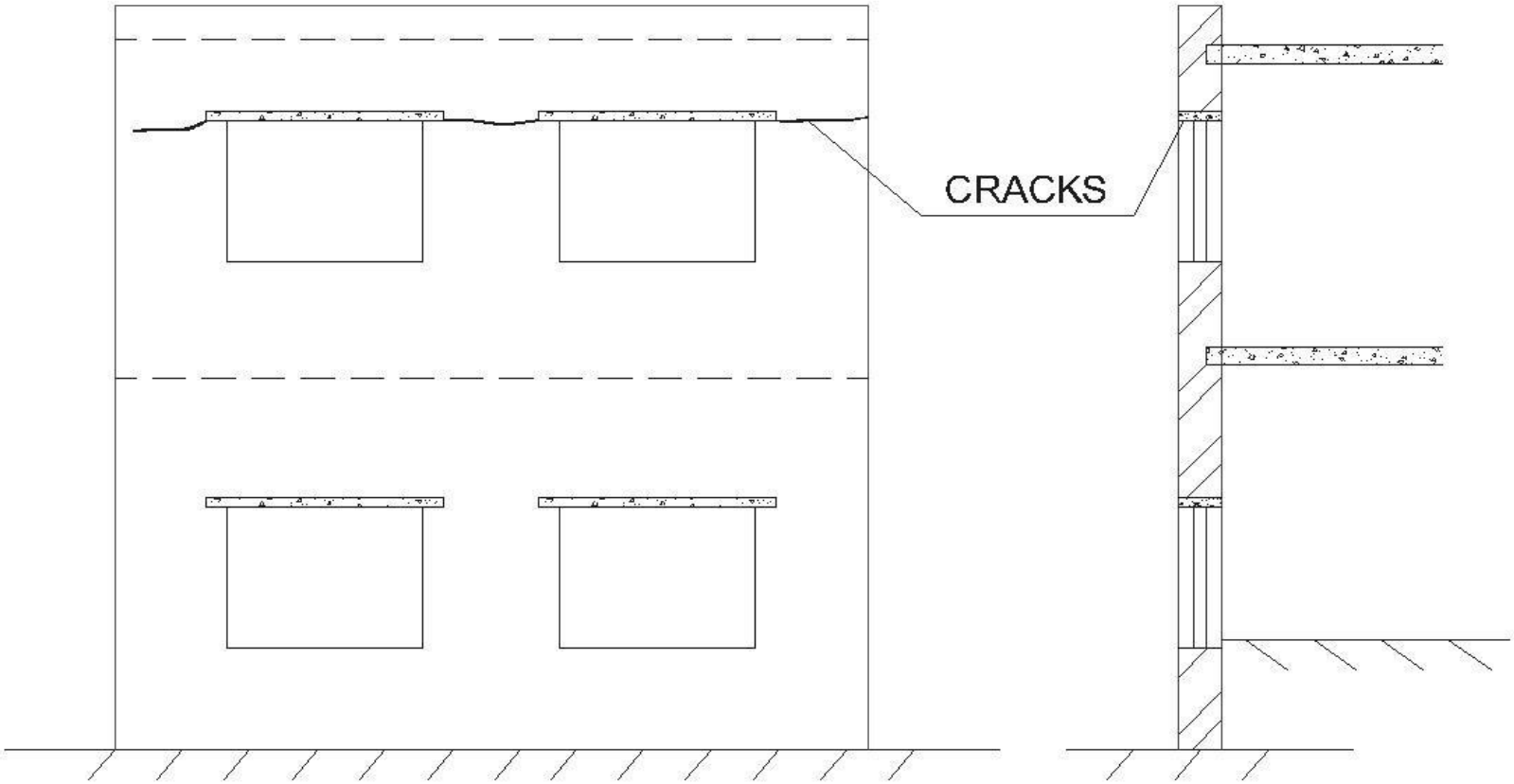
Horizontal Cracks in Brick Panels of a Framed Structure



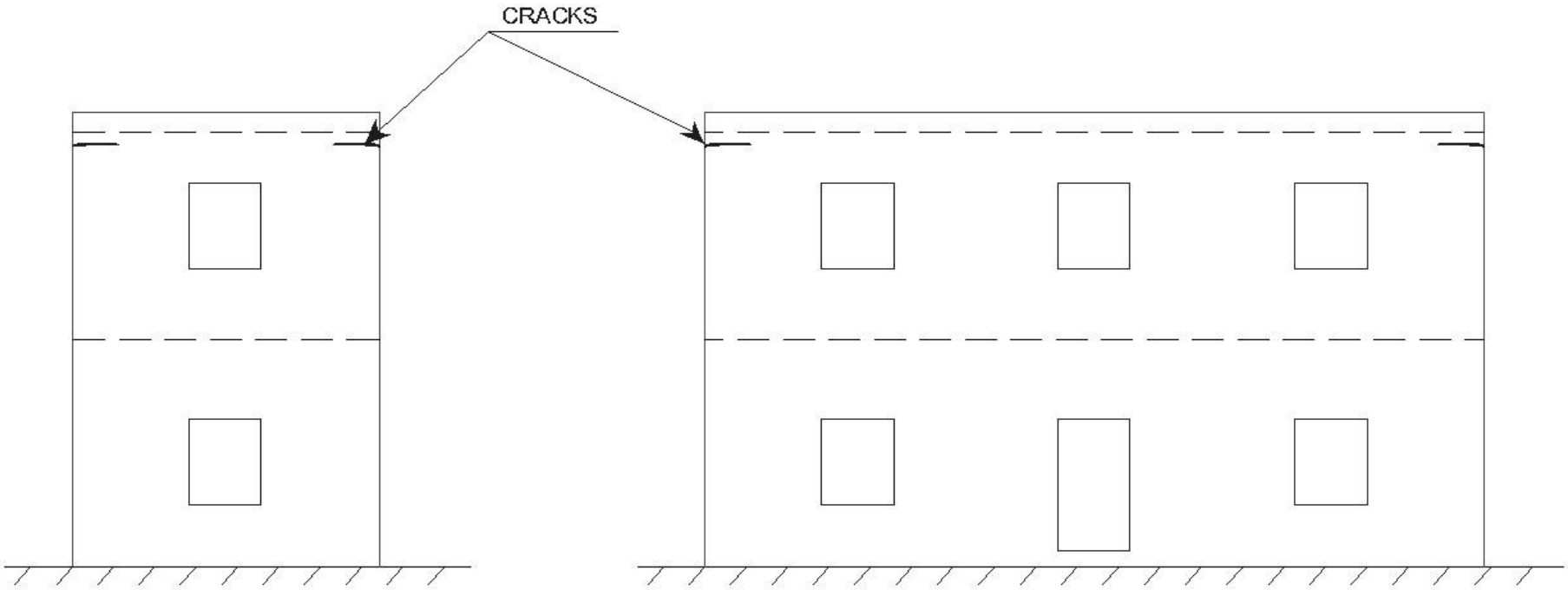
Vertical Cracks in the External Wall around Staircase Opening in a Long Building



Cracks in the External Wall around
RCC Balcony



Horizontal Cracks at Window Lintel Level Due in Top Most Storey



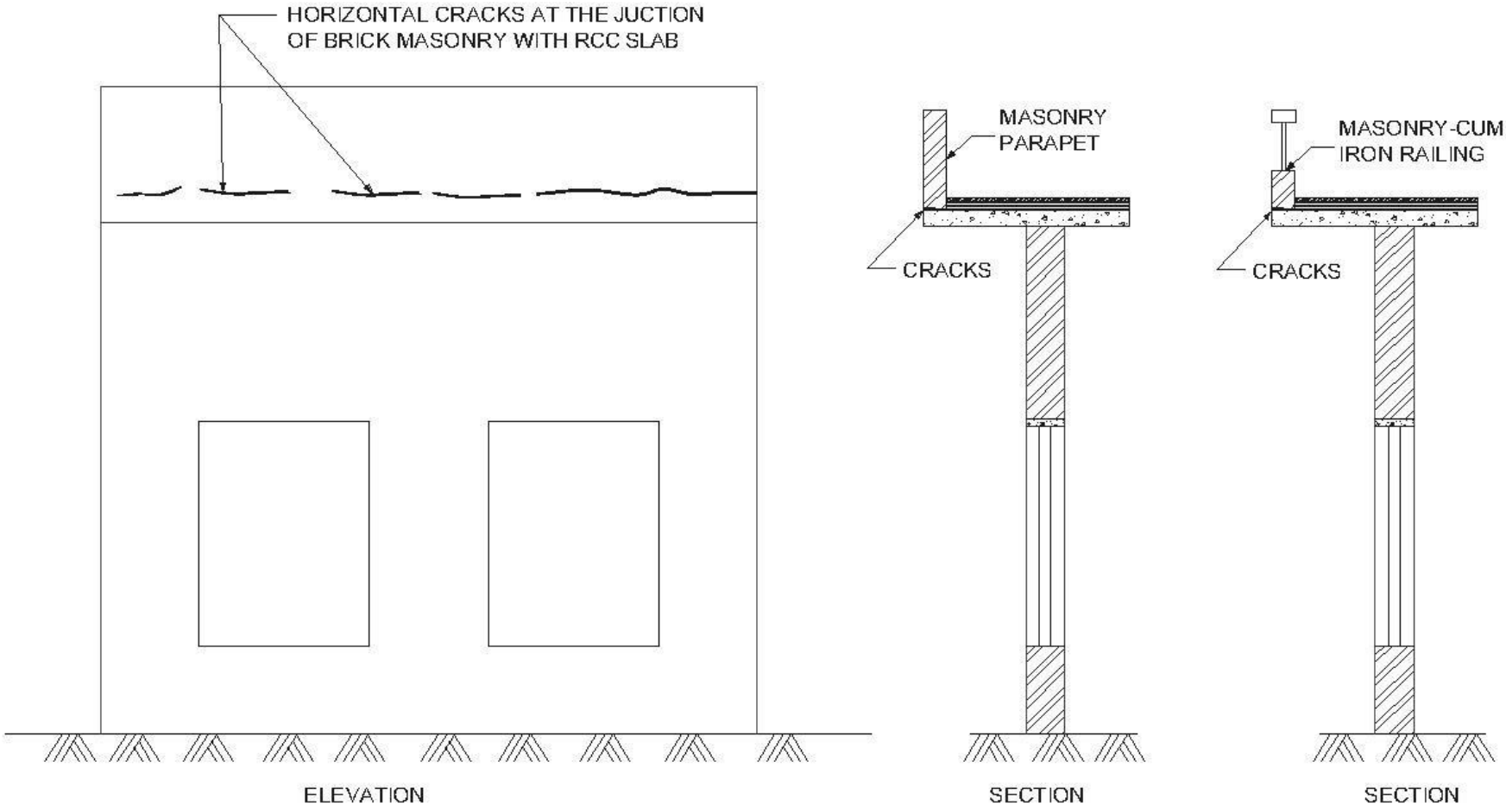
Horizontal Cracks at Corner of Top Storey Below Slab Level Due to Lifting of Corners

Thermal expansion and shrinkage

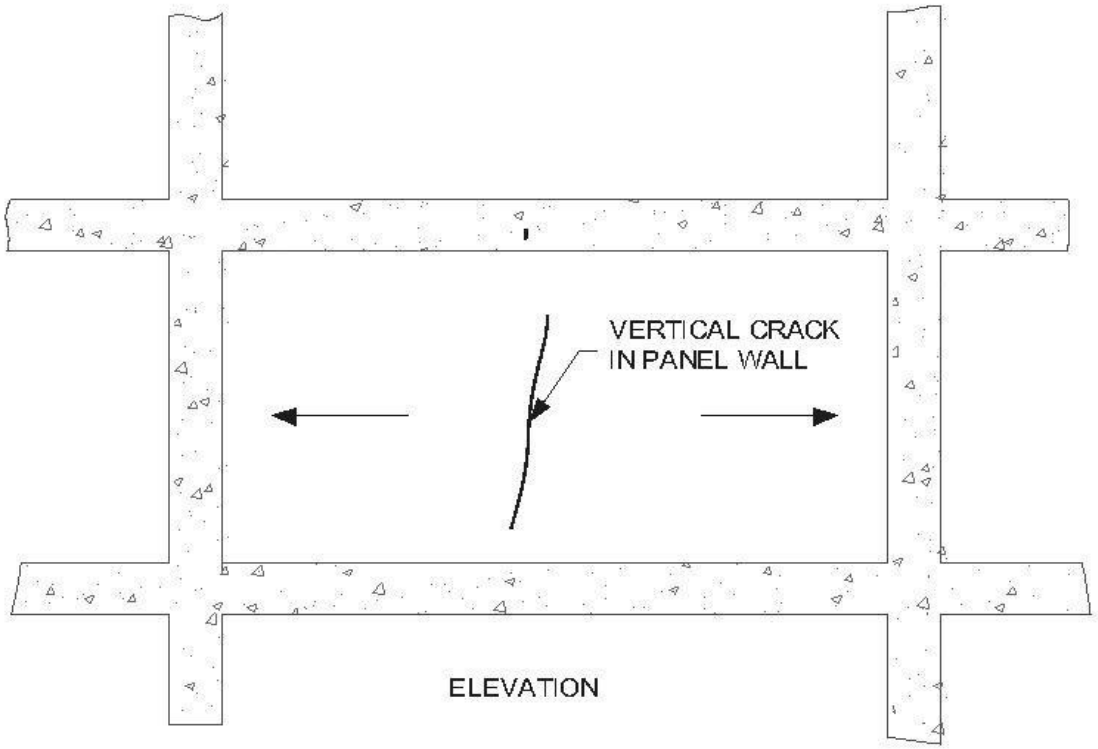
- Assume characteristics compressive strength of concrete -25 MPa
- Modulus of elasticity – $5000x \sqrt{fck}$ N/mm^2
  - $5000 \times \sqrt{25} = 25 \times 10^4$ N/mm^2
- Flexural strength = $0.7 \times \sqrt{fck} = 3.5$ N/mm^2
- Coefficient of thermal expansion of concrete = $10 \times 10^{-6}/°c$
- Diurnal Variation in temperature = 20 °c

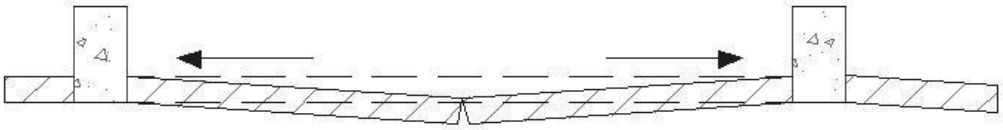
Continue…

- Thermal shrinkage strain = $20 \times 10 \times 10^{-6}$
- Modulus of elasticity = Stress/Strain
- 2.5×10^{4} = Stress/200×10^{-6}
- Tensile Stress = 5.0 N/mm^2
- Tensile Strength of concrete = 3.5 N/mm^2
- Sure to cause crack

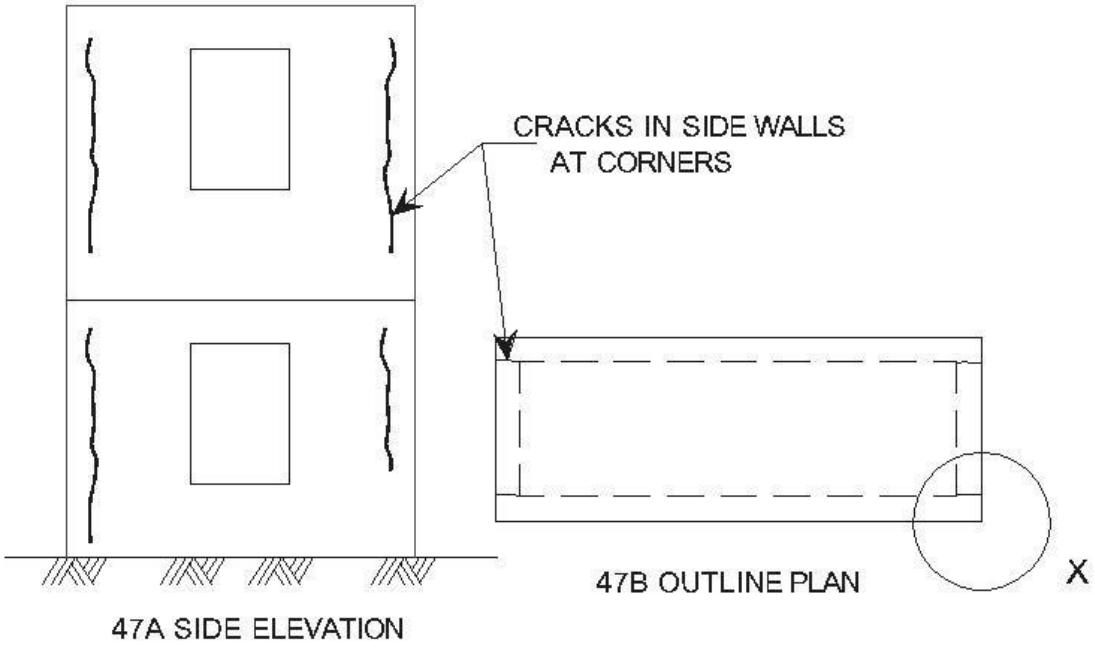


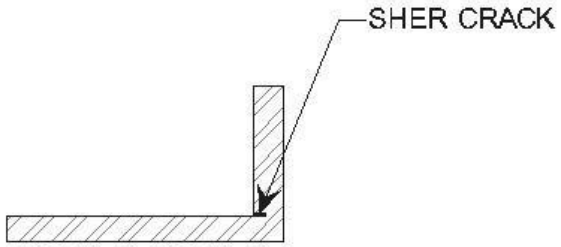
Horizontal Cracks at the Base of Brick Masonary Parapet (or Masonary-cum-Iron Railing)
Supported on a Projection RCC Slab



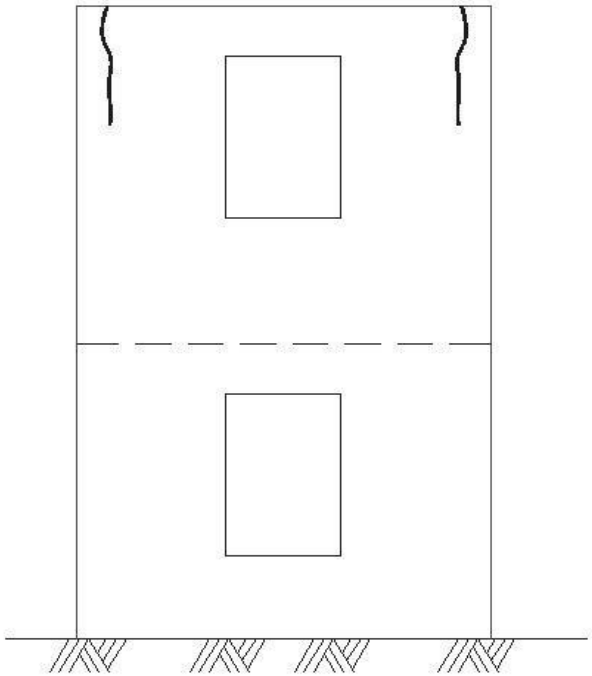


Vertical Cracks in Brick Wall of Framed Structure Due to Expansion of Brickwork

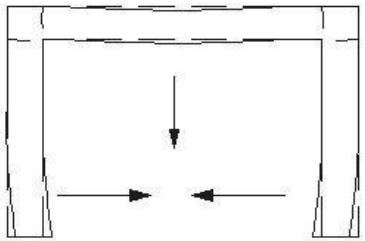




Vertical Cracks at Corner in the in the Side Walls of a Building Due to Thermal Movement



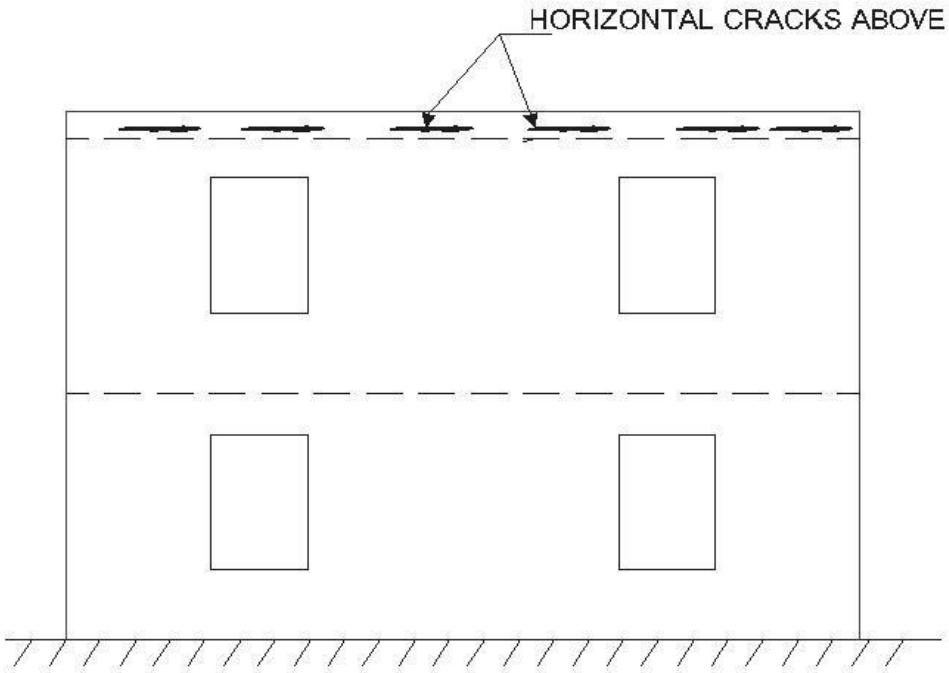
48A ELEVATION



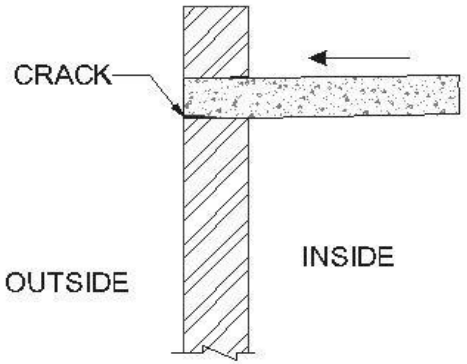
Arrows Indicate Direction of
Pull Of Slab on Walls

48B MECHANISM OF CRACKING

Vertical Cracks at Corner in the in the Top Storey of a Building Due to Drying Shrinkage and Thermal Contraction of Slab



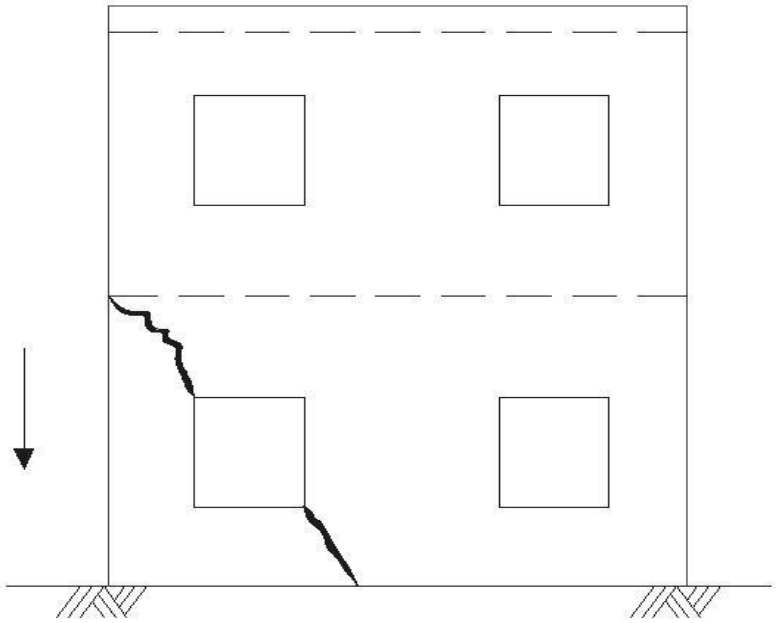
52A ELEVATION



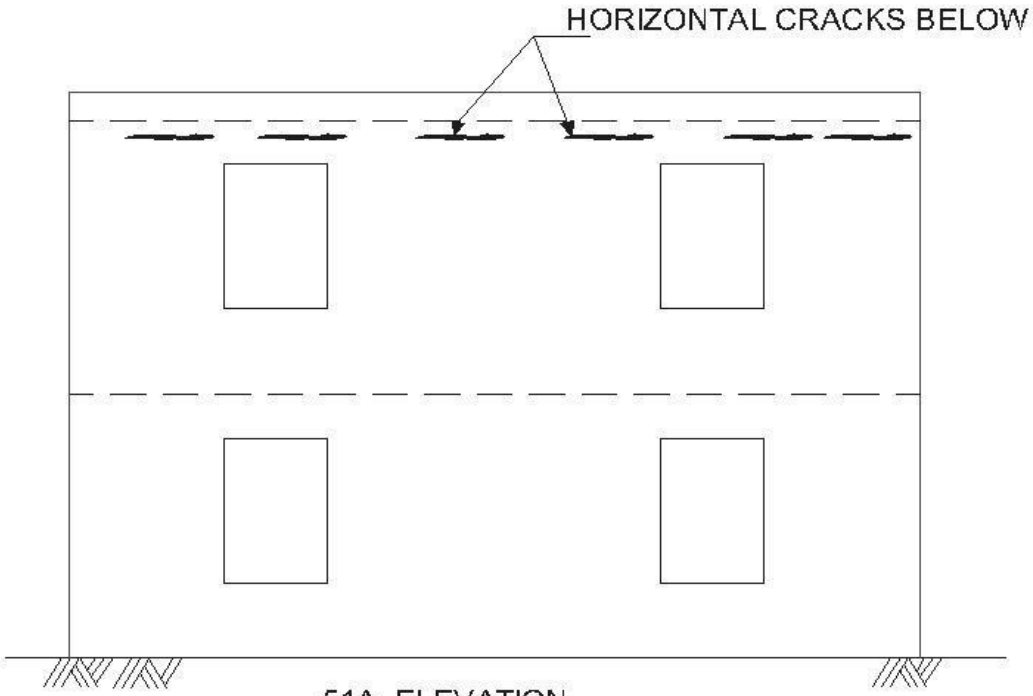
52B SECTION ACROSS THE CRACK

Horizontal Cracks Above the RCC Slab in Top-Most Story Due to Arching and Expansion of Slab

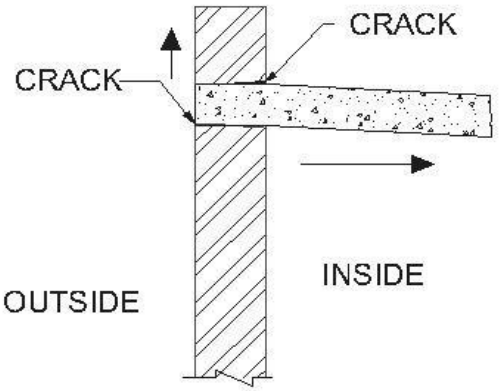
Foundation settlement…



Cracks at the Center of a Building
Due to Foundation Settlement



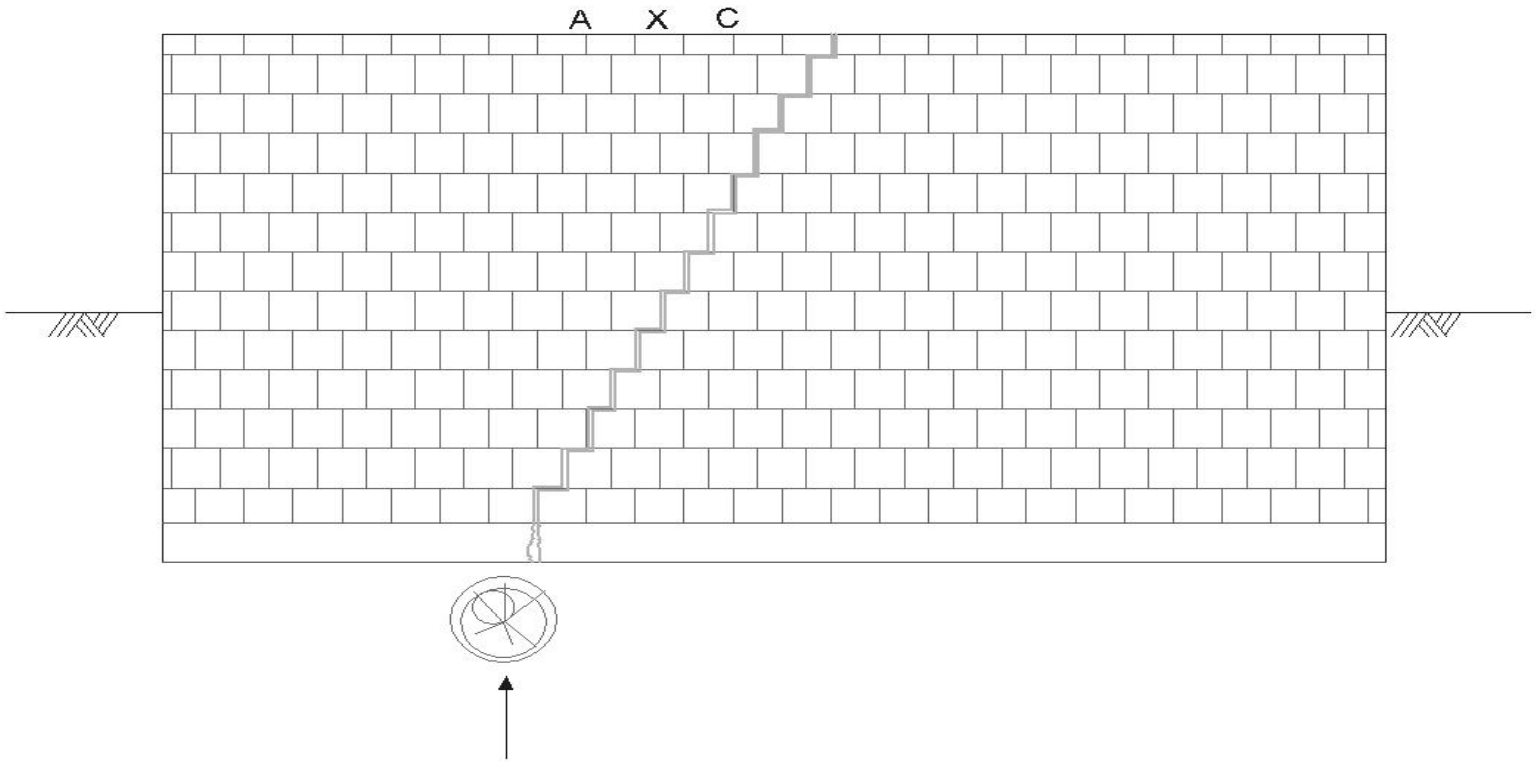
51A ELEVATION



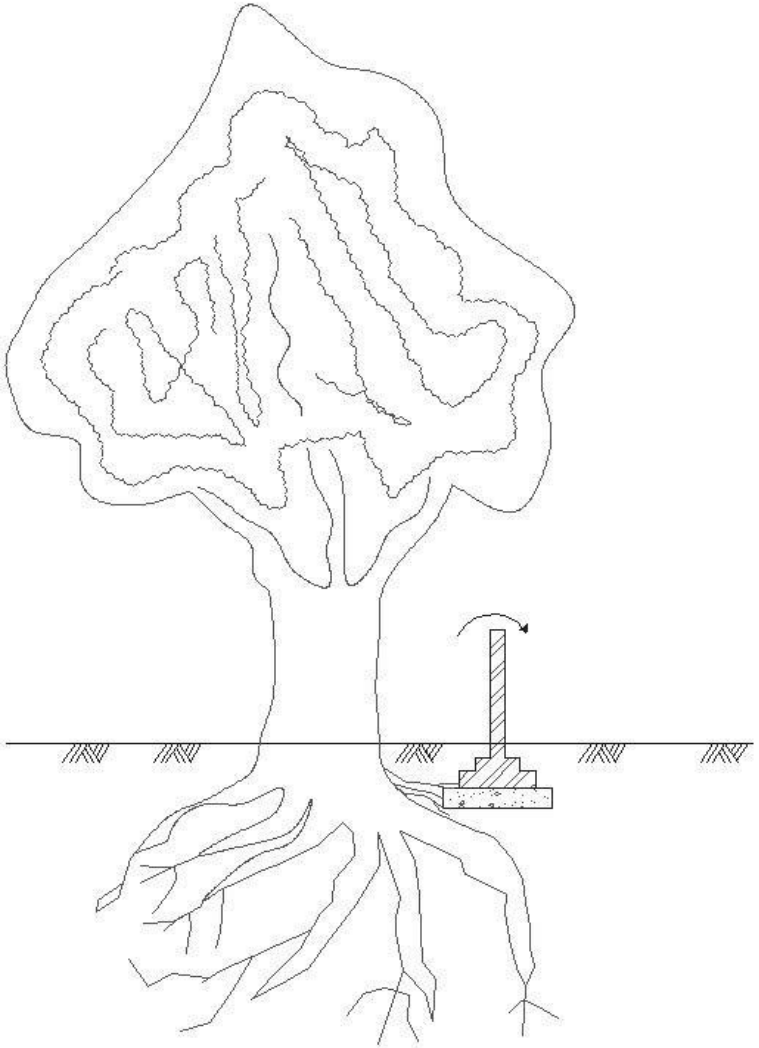
51B SECTION ACROSS THE CRACK

Horizontal Cracks in Top-Most Story Below Slab Due to Shrinkage and Deflection of Slab

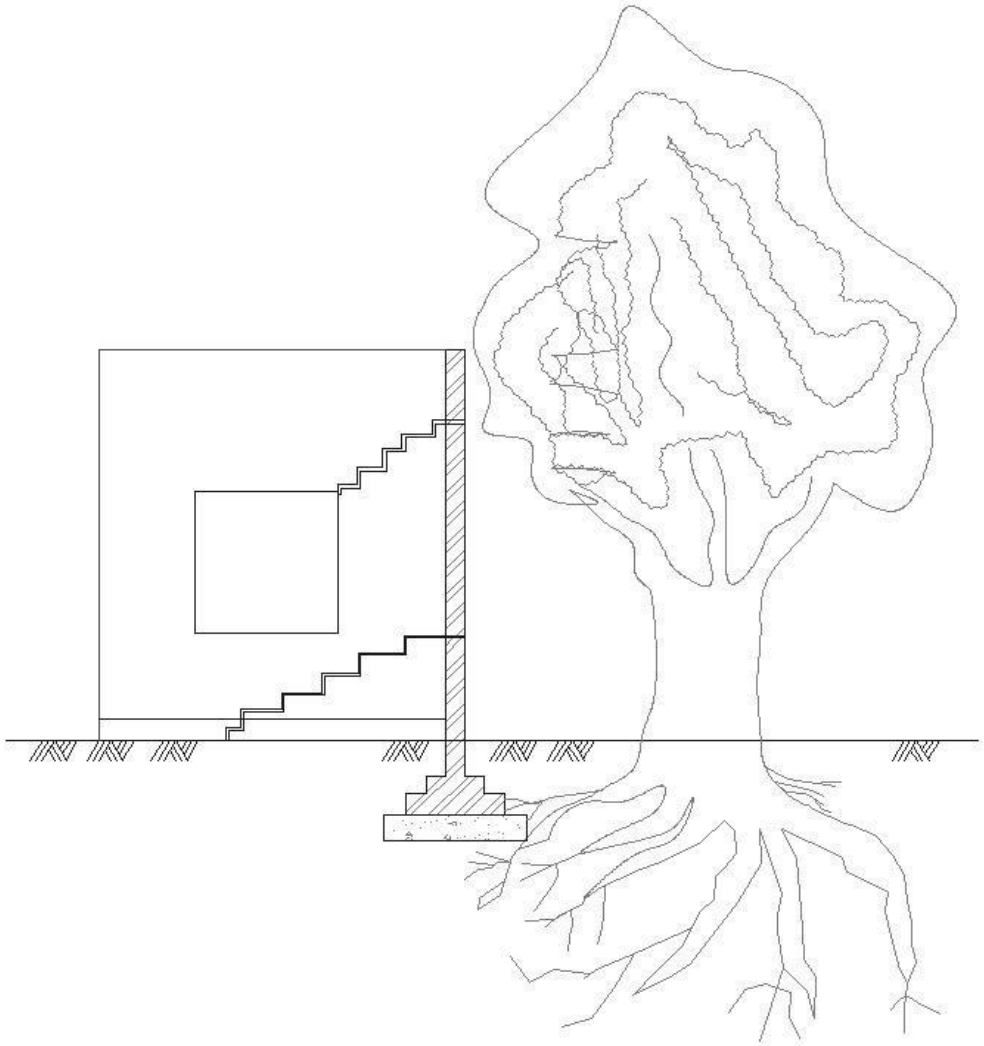
Crack due to vegetation…



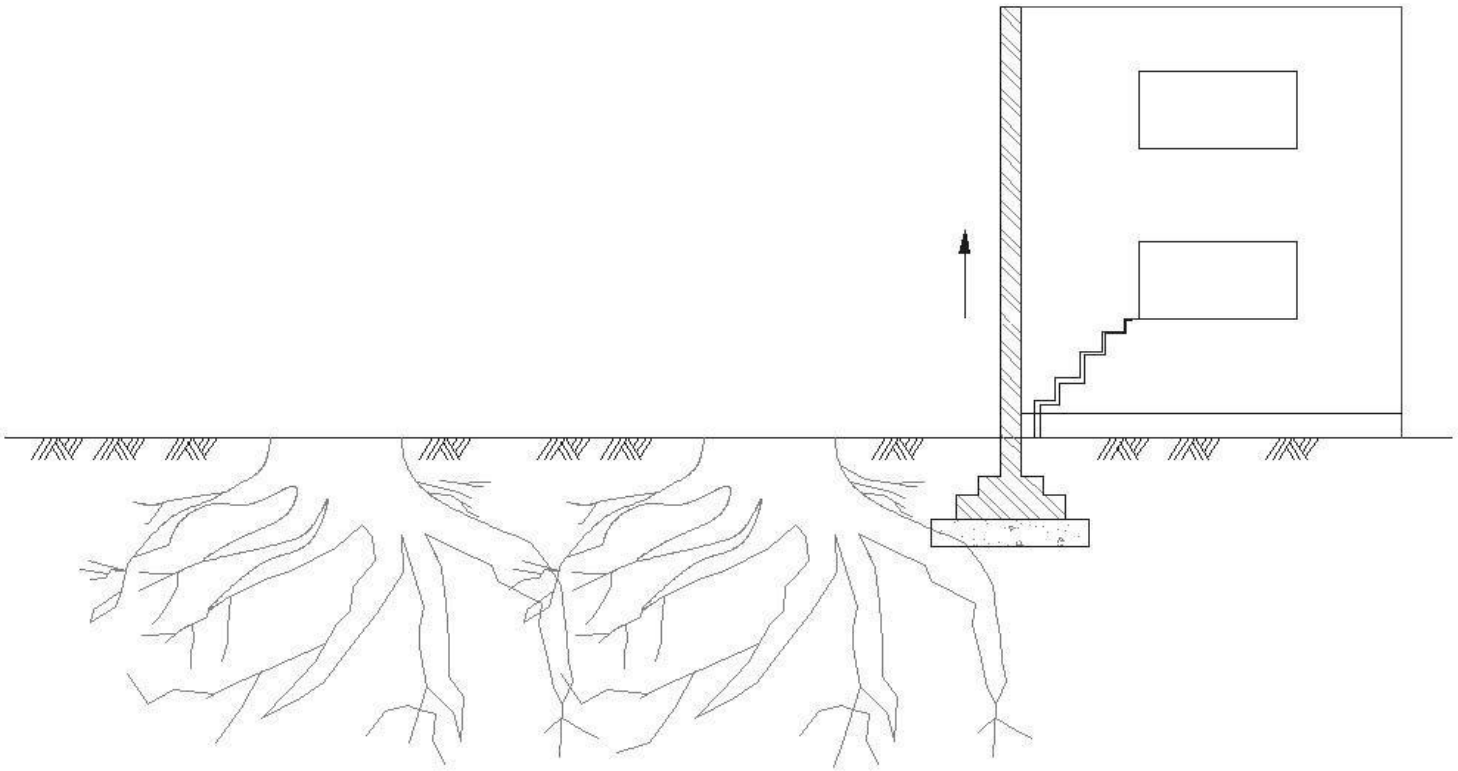
Cracking of a Compund Wall Due to Growing Roots Under Foundation



Roots of Fast Growing Tree Under
The Foundation of Compound Wall May
Topple Down the Wall



Trees Growing Close to a Building on Shrinkable Soil may Cause Crack in the Walls



Cracking Due to Expansion of Soil, if Construction is Taken up Soon After Removal of Tress